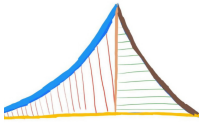
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE EC 2025

A. Formulae

1.1.1 Rank of a Matrix.

$$\text{rank}\,(\mathbf{A}) = \text{number of non-zero rows of } \mathbf{A} \text{ in row echelon form} \tag{1.1.1.1}$$

1.1.2 p-Series Test.

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \iff p > 1 \tag{1.1.2.1}$$

1.1.3 Telescoping Series.

$$\sum_{n=1}^{N} \frac{1}{n\,(n+1)} = \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1} \right) \tag{1.1.3.1}$$

$$= 1 - \frac{1}{N+1} \tag{1.1.3.2}$$

1.1.4 Exponential Series.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{1.1.4.1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1 \tag{1.1.4.2}$$

Solution: Substituting $x = 1$ and removing the term $n = 0$ gives (**??**).

1.1.5 Ratio Test. For a series $\sum a_n$,

$$L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| \tag{1.1.5.1}$$

$$L < 1 \implies \text{the series converges} \tag{1.1.5.2}$$

1.1.6 Expectation.

$$E\,[X] = \sum_{x} x \,\Pr\,(X = x) \tag{1.1.6.1}$$

$$E\,[X_1 + X_2] = E\,[X_1] + E\,[X_2] \tag{1.1.6.2}$$

1.1.7 Marginal PMF.

$$\Pr\,(X = x) = \sum_{y} \Pr\,(X = x, Y = y) \tag{1.1.7.1}$$

1.1.8 Independent Random Variables. If X and Y are independent,

$$\Pr\,(X = x, Y = y) = \Pr\,(X = x)\,\Pr\,(Y = y) \tag{1.1.8.1}$$

1.1.9 PMF of a Sum of Independent Random Variables.

$$p_{X+Y}[k] = \sum_i p_X[i]\, p_Y[k-i] \tag{1.1.9.1}$$

Solution: The event $\{X+Y=k\}$ is the union over i of the disjoint events $\{X=i, Y=k-i\}$. Hence

$$\Pr(X+Y=k) = \sum_i \Pr(X=i, Y=k-i) \tag{1.1.9.2}$$

$$= \sum_i \Pr(X=i)\Pr(Y=k-i) \tag{1.1.9.3}$$

$$= \sum_i p_X[i]\, p_Y[k-i] \tag{1.1.9.4}$$

upon using (**??**).

1.1.10 Uncorrelated Random Variables. For independent, zero-mean, unit-variance a_n and a_m, $n \neq m$,

$$E[a_n a_m] = E[a_n]\, E[a_m] = 0 \tag{1.1.10.1}$$

$$E\left[a_n^2\right] = 1 \tag{1.1.10.2}$$

$$E[a_n a_m] = \delta_{n-m} \tag{1.1.10.3}$$

1.1.11 Gaussian Symmetry. The density of a zero-mean Gaussian random variable N is symmetric, so

$$\Pr(N > c) = \Pr(N < -c) \tag{1.1.11.1}$$

1.1.12 Admittances.

$$Y_R = \frac{1}{R} \tag{1.1.12.1}$$

$$Y_L = \frac{1}{\jmath\omega L} \tag{1.1.12.2}$$

$$Y_C = \jmath\omega C \tag{1.1.12.3}$$

1.1.13 Ohm's Law for Phasors.

$$\underline{I} = Y\underline{V} \tag{1.1.13.1}$$

1.1.14 Polar Form of Phasors.

$$\jmath = 1\angle 90^\circ \tag{1.1.14.1}$$

$$-\jmath = 1\angle -90^\circ \tag{1.1.14.2}$$

$$(a\angle\theta_1)\,(b\angle\theta_2) = ab\angle(\theta_1+\theta_2) \tag{1.1.14.3}$$

1.1.15 Series RLC Branch.

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \tag{1.1.15.1}$$

$$\omega L = \frac{1}{\omega C} \tag{1.1.15.2}$$

$$C = \frac{1}{\omega^2 L} \tag{1.1.15.3}$$

1.1.16 Z-Parameters of a Two-Port.

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \tag{1.1.16.1}$$

$$z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0} \tag{1.1.16.2}$$

$$z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0} \tag{1.1.16.3}$$

1.1.17 Parallel Resistors.

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \tag{1.1.17.1}$$

1.1.18 LTI Systems. For a system $y[n] = T\{x[n]\}$,

$$\delta[n] * h[n] = h[n] \tag{1.1.18.1}$$

$$T\{ax_1 + bx_2\} = aT\{x_1\} + bT\{x_2\} \quad \text{(linear)} \tag{1.1.18.2}$$

$$T\{0\} = 0 \quad \text{(a linear system)} \tag{1.1.18.3}$$

$$T\{x[n-k]\} = y[n-k] \quad \text{(time invariant)} \tag{1.1.18.4}$$

1.1.19 Fourier Transform.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-\jmath\omega t}\, dt \tag{1.1.19.1}$$

1.1.20 Magnitude Bound of the Fourier Transform.

$$|F(\omega)| \le \int_{-\infty}^{\infty} |f(t)|\, dt \tag{1.1.20.1}$$

Solution: Using $\left|\int g\right| \le \int |g|$ and $|e^{-\jmath\omega t}| = 1$ in (**??**),

$$|F(\omega)| \le \int_{-\infty}^{\infty} |f(t)| \left|e^{-\jmath\omega t}\right| dt \tag{1.1.20.2}$$

$$= \int_{-\infty}^{\infty} |f(t)|\, dt \tag{1.1.20.3}$$

1.1.21 Transform of $e^{-|t|}$.

$$e^{-|t|} \longleftrightarrow \frac{2}{1+\omega^2} \tag{1.1.21.1}$$

Solution: From (**??**),

$$\int_{-\infty}^{\infty} e^{-|t|} e^{-\jmath\omega t}\, dt = \int_{-\infty}^{0} e^{t(1-\jmath\omega)}\, dt + \int_{0}^{\infty} e^{-t(1+\jmath\omega)}\, dt \tag{1.1.21.2}$$

$$= \frac{1}{1-\jmath\omega} + \frac{1}{1+\jmath\omega} \tag{1.1.21.3}$$

$$= \frac{2}{1+\omega^2} \tag{1.1.21.4}$$

1.1.22 Fourier Series and Time Scaling. A periodic signal with fundamental period T_0 has

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{\jmath 2\pi k t/T_0} \tag{1.1.22.1}$$

and, for $\alpha > 0$,

$$f(\alpha t) = \sum_{k=-\infty}^{\infty} c_k e^{\jmath 2\pi k \alpha t/T_0} \tag{1.1.22.2}$$

$$T_y = \frac{T_0}{\alpha} \tag{1.1.22.3}$$

Solution: Replacing t by αt in (**??**) yields (**??**), with the same coefficients c_k. For $y(t) = f(\alpha t)$,

$$y\left(t + \frac{T_0}{\alpha}\right) = f(\alpha t + T_0) \tag{1.1.22.4}$$

$$= f(\alpha t) = y(t) \tag{1.1.22.5}$$

yielding (**??**).

1.1.23 Impulse Sampling. With $\omega_s = \frac{2\pi}{T_s}$,

$$\sum_n \delta(t - \tau - nT_s) = \frac{1}{T_s} \sum_k e^{\jmath k\omega_s (t-\tau)} \tag{1.1.23.1}$$

$$Y(\omega) = \frac{1}{T_s} \sum_k e^{-\jmath k\omega_s \tau} F(\omega - k\omega_s) \tag{1.1.23.2}$$

where $y(t) = f(t) \sum_n \delta(t - \tau - nT_s)$. **Solution:** Multiplying $f(t)$ by (**??**) and using that $f(t) e^{\jmath k\omega_s t}$ has transform $F(\omega - k\omega_s)$,

$$y(t) = \frac{1}{T_s} \sum_k e^{-\jmath k\omega_s \tau} f(t) e^{\jmath k\omega_s t} \tag{1.1.23.3}$$

$$Y(\omega) = \frac{1}{T_s} \sum_k e^{-\jmath k\omega_s \tau} F(\omega - k\omega_s) \tag{1.1.23.4}$$

1.1.24 Ideal Lowpass Reconstruction.

$$H(\omega) = T_s, \quad |\omega| < \omega_c \tag{1.1.24.1}$$

$$\omega_s > 2\omega_c \iff T_s < \frac{\pi}{\omega_c} \tag{1.1.24.2}$$

1.1.25 z-Transform and DTFT.

$$X(z) = \sum_n x[n] z^{-n} \tag{1.1.25.1}$$

$$X(e^{\jmath\omega}) = \sum_n x[n] e^{-\jmath\omega n} \tag{1.1.25.2}$$

The DTFT exists if the region of convergence (ROC) contains the unit circle. The ROC contains no poles but may contain zeros. For $x[n] = \alpha\delta[n]$, $X(z) = \alpha$ with the ROC the entire z-plane.

1.1.26 Characteristic Equation of a Unity-Feedback Loop.

$$1 + L(s) = 0 \tag{1.1.26.1}$$

1.1.27 Stability of a Quadratic Polynomial.

$$a_2 s^2 + a_1 s + a_0 \text{ is stable } \iff a_2, a_1, a_0 \text{ have the same sign} \tag{1.1.27.1}$$

1.1.28 Roots of $s^3 + c$. For $c > 0$ and $k = 0, 1, 2$,

$$s^3 + c = 0 \implies s = c^{1/3} e^{\jmath(\pi + 2\pi k)/3} \tag{1.1.28.1}$$

1.1.29 Root Locus. A point s_0 lies on the root locus of $1 + KG(s) = 0$ for some $K \geq 0$ iff

$$G(s_0) = -\frac{1}{K} \tag{1.1.29.1}$$

i.e. $G(s_0)$ is real and negative. **Solution:** At a closed-loop pole $1 + KG(s_0) = 0$, hence

$$G(s_0) = -\frac{1}{K} \tag{1.1.29.2}$$

$$K = \frac{1}{|G(s_0)|} \tag{1.1.29.3}$$

1.1.30 Mason's Gain Formula.

$$T = \frac{\sum_k P_k \Delta_k}{\Delta} \tag{1.1.30.1}$$

$$\Delta = 1 - \sum_i L_i + \cdots \tag{1.1.30.2}$$

where P_k are the forward-path gains, L_i the loop gains and Δ_k is Δ with the loops touching path k removed.

1.1.31 State Equation. For $\dot{x}_i = \lambda_i x_i + b_i u$,

$$x_i(t) = e^{\lambda_i t} x_i(0) + b_i \int_0^t e^{\lambda_i (t-\sigma)} u(\sigma)\, d\sigma \tag{1.1.31.1}$$

$$|x_i(t)| \leq |x_i(0)| + \frac{|b_i| U}{|\lambda_i|} \quad \text{if } \lambda_i < 0, |u| \leq U \tag{1.1.31.2}$$

$$x_i(t) = \frac{b_i}{\lambda_i}\left(e^{\lambda_i t} - 1\right) \quad \text{if } \lambda_i > 0, u \equiv 1, x_i(0) = 0 \tag{1.1.31.3}$$

$$x_i(t) = b_i t \quad \text{if } \lambda_i = 0, u \equiv 1, x_i(0) = 0 \tag{1.1.31.4}$$

Solution: Putting $u \equiv 1$ and $x_i(0) = 0$ in (**??**),

$$x_i(t) = b_i \int_0^t e^{\lambda_i (t-\sigma)}\, d\sigma \tag{1.1.31.5}$$

$$= \frac{b_i}{\lambda_i}\left(e^{\lambda_i t} - 1\right), \quad \lambda_i \neq 0 \tag{1.1.31.6}$$

$$= b_i t, \quad \lambda_i = 0 \tag{1.1.31.7}$$

which grows without bound for $\lambda_i \geq 0$ and $b_i \neq 0$.

1.1.32 Majority Function.

$$Y_{\text{maj}} = AB + AC + BC \tag{1.1.32.1}$$

1.1.33 XOR with a Constant.

$$Y \oplus 0 = Y \tag{1.1.33.1}$$

$$Y \oplus 1 = \overline{Y} \tag{1.1.33.2}$$

1.1.34 Two's Complement Subtraction.

$$X + \overline{Y} + 1 \equiv X - Y \pmod{2^n} \tag{1.1.34.1}$$

Solution: For n-bit numbers $\overline{Y} = (2^n - 1) - Y$, so

$$X + \overline{Y} + 1 = X + (2^n - 1) - Y + 1 \tag{1.1.34.2}$$

$$= X - Y + 2^n \tag{1.1.34.3}$$

$$\equiv X - Y \pmod{2^n} \tag{1.1.34.4}$$

1.1.35 D Flip-Flop and Setup Constraint.

$$Q^+ = D \tag{1.1.35.1}$$

$$T_{\text{clk}} \geq t_{cq} + t_{comb} + t_{setup} \tag{1.1.35.2}$$

$$f_{\max} = \frac{1}{T_{\text{clk,min}}} \tag{1.1.35.3}$$

1.1.36 Frequency Modulation.

$$f_i(t) = \frac{1}{2\pi}\frac{d\phi(t)}{dt} \tag{1.1.36.1}$$

$$\Delta f_{\text{peak}} = \max_t |f_i(t) - f_c| \tag{1.1.36.2}$$

1.1.37 Envelope Detection.

$$\text{envelope (DSB-SC)} = A_c|m(t)| \tag{1.1.37.1}$$

$$\text{envelope (AM)} = A_c|1 + \mu m(t)| \tag{1.1.37.2}$$

1.1.38 Maximum-Likelihood Detection. For equal priors and equal-variance Gaussian noise, the detector chooses the nearest symbol

$$\hat{S} = \arg\min_s |y - s| \tag{1.1.38.1}$$

1.1.39 Ideal ADC.

$$\text{SNR}_{\max} = 6.02N + 1.76 \text{ dB} \tag{1.1.39.1}$$

$$R_b = f_s N \tag{1.1.39.2}$$

1.1.40 Inverting Op-Amp Filter.

$$H(s) = -\frac{Z_f(s)}{Z_i(s)} \tag{1.1.40.1}$$

A factor $\frac{1}{1+sRC}$ gives a corner frequency

$$\omega = \frac{1}{RC} \tag{1.1.40.2}$$

1.1.41 Entropy.

$$H(X) = -\sum_x p(x)\log_2 p(x) \tag{1.1.41.1}$$

1.1.42 Mutual Information.

$$I(X;Y) = \sum_{x,y} p(x,y)\log_2 \frac{p(x,y)}{p(x)p(y)} \tag{1.1.42.1}$$

$$I(X;Y) = 0 \quad \text{if } X, Y \text{ are independent} \tag{1.1.42.2}$$

Solution: For independent X, Y, $p(x,y) = p(x)p(y)$ by (**??**), so every logarithm in (**??**) is $\log_2 1 = 0$:

$$I(X;Y) = \sum_{x,y} p(x,y)\log_2 1 \tag{1.1.42.3}$$

$$= 0 \tag{1.1.42.4}$$

1.1.43 Minimum Distance of a Binary Linear Code. With w the Hamming weight,

$$d_{\min} = \min_{\mathbf{c}\neq\mathbf{0}} w(\mathbf{c}) \tag{1.1.43.1}$$

1.1.44 Faraday's Law.

$$|\mathcal{E}| = \left|\frac{d\Phi_B}{dt}\right| = A\left|\frac{dB}{dt}\right| \tag{1.1.44.1}$$

$$I = \frac{|\mathcal{E}|}{R_{\text{total}}} \tag{1.1.44.2}$$

1.1.45 Current Density.

$$J = \sigma E \tag{1.1.45.1}$$

$$I = JA \tag{1.1.45.2}$$

1.1.46 Surface Charge.

$$Q = \iint \rho_s \, dA \tag{1.1.46.1}$$

1.1.47 Ideal Diode and Phase-to-Time Conversion. An ideal diode is forward biased iff the anode voltage exceeds the cathode voltage. A phase interval $\Delta\theta$ at angular frequency ω lasts

$$\Delta t = \frac{\Delta\theta}{\omega} \tag{1.1.47.1}$$

1.1.48 Second Derivative Test.

$$f'(x_0) = 0, f''(x_0) > 0 \implies \text{local minimum} \tag{1.1.48.1}$$

$$f'(x_0) = 0, f''(x_0) < 0 \implies \text{local maximum} \tag{1.1.48.2}$$

1.1.49 Bounds on an Integral.

$$m \le f(x) \le M \text{ on } [a,b] \implies m(b-a) \le \int_a^b f(x)\, dx \le M(b-a) \tag{1.1.49.1}$$

1.1.50 Euler–Cauchy Equation. For $t^2y'' + aty' + by = 0$ with $y = t^m$,

$$m(m-1) + am + b = 0 \tag{1.1.50.1}$$

Solution: Substituting $y = t^m$, $y' = mt^{m-1}$ and $y'' = m(m-1)t^{m-2}$,

$$t^2y'' + aty' + by = t^m\left[m(m-1) + am + b\right] \tag{1.1.50.2}$$

$$= 0 \tag{1.1.50.3}$$

yielding (**??**).

1.1.51 Least Squares. The scalar x minimizing $\|x\mathbf{a} - \mathbf{b}\|$ is

$$x^* = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} \tag{1.1.51.1}$$

Solution: Let $J(x) = \|x\mathbf{a} - \mathbf{b}\|^2$. Then

$$J(x) = x^2\mathbf{a}^\top\mathbf{a} - 2x\mathbf{a}^\top\mathbf{b} + \mathbf{b}^\top\mathbf{b} \tag{1.1.51.2}$$

$$\frac{dJ}{dx} = 2x\mathbf{a}^\top\mathbf{a} - 2\mathbf{a}^\top\mathbf{b} = 0 \tag{1.1.51.3}$$

$$x^* = \frac{\mathbf{a}^\top\mathbf{b}}{\mathbf{a}^\top\mathbf{a}} \tag{1.1.51.4}$$

1.1.52 Cauchy's Theorem. For a closed counter-clockwise contour C,

$$\oint_C f(z)\,dz = 0 \quad \text{if } f \text{ is analytic inside and on } C \tag{1.1.52.1}$$

1.1.53 Integral of z^n on the Unit Circle.

$$\oint_{|z|=1} z^n\,dz = \begin{cases} 2\pi\jmath, & n = -1 \\ 0, & n \neq -1 \end{cases} \tag{1.1.53.1}$$

Solution: Putting $z = e^{\jmath\theta}$ and $dz = \jmath e^{\jmath\theta}d\theta$,

$$\oint_{|z|=1} z^n\,dz = \jmath\int_0^{2\pi} e^{\jmath(n+1)\theta}\,d\theta \tag{1.1.53.2}$$

$$= \begin{cases} 2\pi\jmath, & n = -1 \\ 0, & n \neq -1 \end{cases} \tag{1.1.53.3}$$

1.1.54 Poles of $\sec z$. The poles of $\sec z = \frac{1}{\cos z}$ are where $\cos z = 0$:

$$z = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z} \tag{1.1.54.1}$$

B. Problems

1.2.1 Consider the matrix $\mathbf{A}$ below:

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{pmatrix} \tag{1.2.1.1}$$

For which of the following combinations of α, β and γ is the rank of $\mathbf{A}$ at least three?
(i) $\alpha = 0$ and $\beta = \gamma \neq 0$.
(ii) $\alpha = \beta = \gamma = 0$.
(iii) $\beta = \gamma = 0$ and $\alpha \neq 0$.
(iv) $\alpha = \beta = \gamma \neq 0$. (EC 2025)

a) Only (i), (iii), and (iv)
b) Only (iv)
c) Only (ii)
d) Only (i) and (iii)

Solution: From (**??**), the rank is the number of non-zero rows in the row echelon form. For case (i), $\alpha = 0$ and $\beta = \gamma \neq 0$, rows 3 and 4 are equal,

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 0 & \beta \\ 0 & 0 & 0 & \beta \end{pmatrix} \tag{1.2.1.2}$$

$$\xrightarrow{R_4 \to R_4 - R_3} \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 0 & \beta \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.2.1.3}$$

$$\implies \text{rank}\,(\mathbf{A}) = 3 \tag{1.2.1.4}$$

For case (ii), $\alpha = \beta = \gamma = 0$,

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.2.1.5}$$

$$\implies \text{rank}\,(\mathbf{A}) = 2 \tag{1.2.1.6}$$

For case (iii), $\beta = \gamma = 0$ and $\alpha \neq 0$,

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.2.1.7}$$

$$\implies \text{rank}\,(\mathbf{A}) = 3 \tag{1.2.1.8}$$

For case (iv), $\alpha = \beta = \gamma \neq 0$,

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \alpha \\ 0 & 0 & 0 & \alpha \end{pmatrix} \tag{1.2.1.9}$$

$$\implies \text{rank}\,(\mathbf{A}) = 4 \tag{1.2.1.10}$$

Hence, only (i), (iii) and (iv) give rank at least three. The following code verifies the solution in Fig. **??**.

codes/Rank_of_Matrix.py

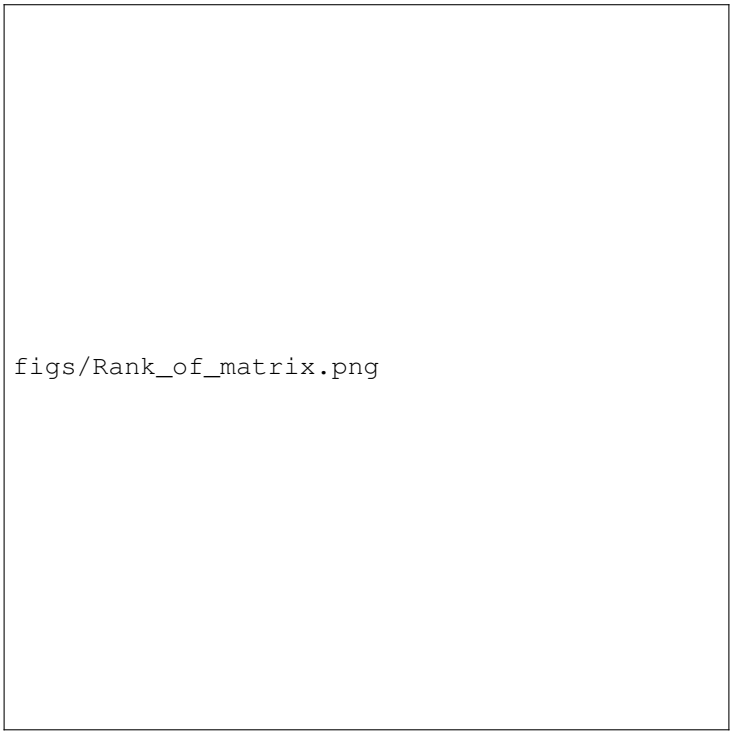


Fig. 1.2.1.1: Rank of $\mathbf{A}$ for the four cases

1.2.2 Consider the following series:

$$\text{(i)} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}, \quad \text{(ii)} \sum_{n=1}^{\infty} \frac{1}{n\,(n+1)}, \quad \text{(iii)} \sum_{n=1}^{\infty} \frac{1}{n!} \tag{1.2.2.1}$$

Choose the correct option. (EC 2025)

a) Only (ii) converges
b) Only (ii) and (iii) converge
c) Only (iii) converges
d) All three converge

Solution: Series (i) is a p-series with $p = \frac{1}{2}$. From (**??**),

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} n^{-\frac{1}{2}} \tag{1.2.2.2}$$

$$p = \frac{1}{2} \leq 1 \tag{1.2.2.3}$$

$$\implies \text{(i) diverges} \tag{1.2.2.4}$$

Series (ii) telescopes. From (**??**),

$$\sum_{n=1}^{N} \frac{1}{n\,(n+1)} = 1 - \frac{1}{N+1} \tag{1.2.2.5}$$

$$\lim_{N\to\infty} \left(1 - \frac{1}{N+1}\right) = 1 \tag{1.2.2.6}$$

For series (iii), from (**??**),

$$\frac{a_{n+1}}{a_n} = \frac{n!}{(n+1)!} \tag{1.2.2.7}$$

$$= \frac{1}{n+1} \tag{1.2.2.8}$$

$$\lim_{n\to\infty} \frac{1}{n+1} = 0 < 1 \tag{1.2.2.9}$$

and from (**??**),

$$\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1 \tag{1.2.2.10}$$

Hence, only (ii) and (iii) converge. The following code plots the partial sums in Fig. **??**.

codes/Series_Convergence.py

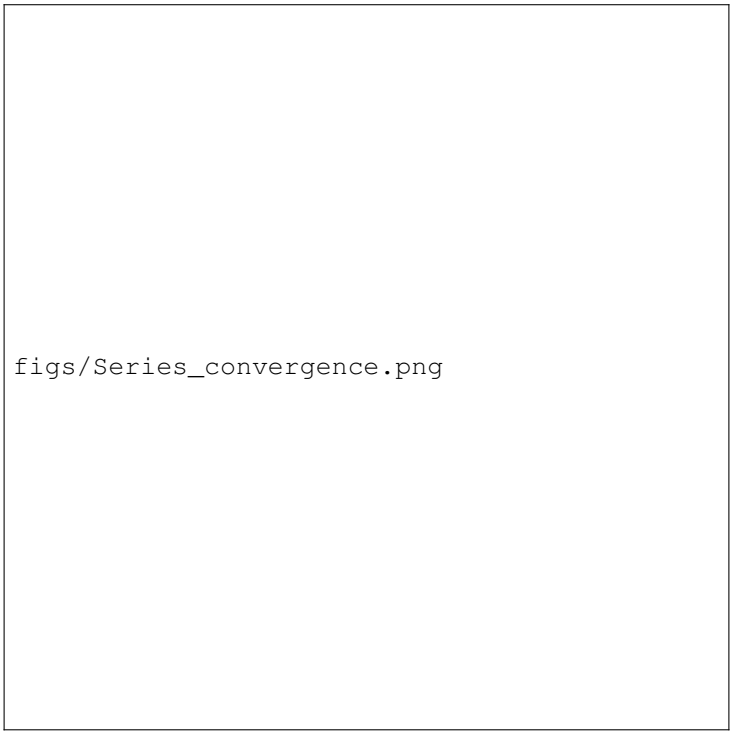


Fig. 1.2.2.1: Partial sums of the three series

1.2.3 A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours? (EC 2025)

a) $\frac{2}{3}$ b) $\frac{1}{3}$ c) $\frac{1}{2}$ d) 1

Solution: Choosing two balls at random is the same as selecting each ball independently with probability $\frac{1}{2}$ and keeping the outcomes with exactly two balls selected. Let B be the selection indicator of one ball, with

$$p_B[0] = p_B[1] = \frac{1}{2} \tag{1.2.3.1}$$

The number X of selected red balls is the sum of two independent indicators. From (**??**),

$$p_X[0] = p_B[0]\, p_B[0] = \frac{1}{4} \tag{1.2.3.2}$$

$$p_X[1] = p_B[0]\, p_B[1] + p_B[1]\, p_B[0] = \frac{1}{2} \tag{1.2.3.3}$$

$$p_X[2] = p_B[1]\, p_B[1] = \frac{1}{4} \tag{1.2.3.4}$$

The number Y of selected blue balls has $p_Y = p_X$. For $S = X + Y$, from (**??**),

$$p_S[2] = p_X[0]\,p_Y[2] + p_X[1]\,p_Y[1] + p_X[2]\,p_Y[0] \tag{1.2.3.5}$$

$$= \frac{1}{16} + \frac{1}{4} + \frac{1}{16} \tag{1.2.3.6}$$

$$= \frac{3}{8} \tag{1.2.3.7}$$

The two balls have different colours when $X = 1$ and $Y = 1$. From (**??**),

$$p_X[1]\,p_Y[1] = \frac{1}{4} \tag{1.2.3.8}$$

$$\Pr(\text{different colours}) = \frac{p_X[1]\,p_Y[1]}{p_S[2]} \tag{1.2.3.9}$$

$$= \frac{\frac{1}{4}}{\frac{3}{8}} \tag{1.2.3.10}$$

$$= \frac{2}{3} \tag{1.2.3.11}$$

The following code simulates the result in Fig. **??**.

codes/Different_Colours.py

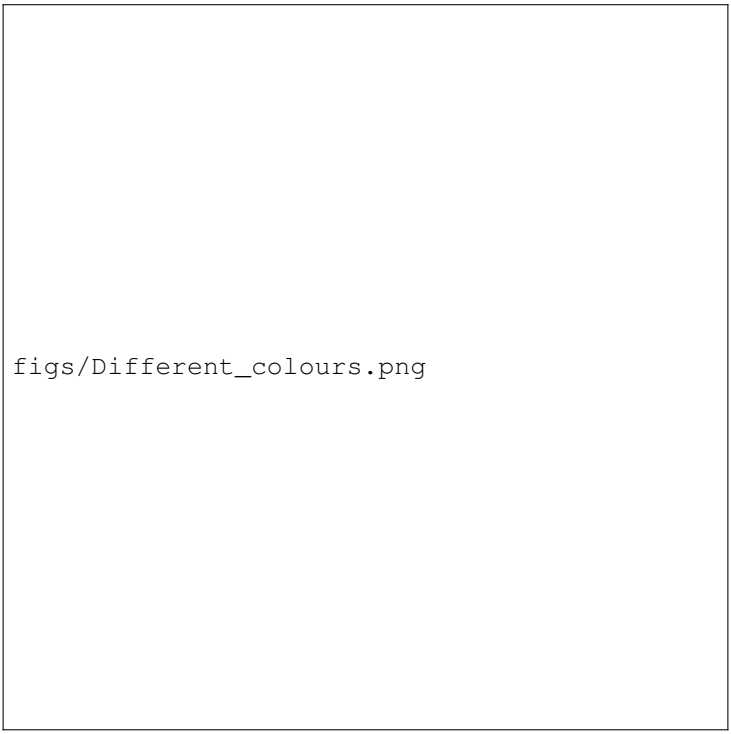


Fig. 1.2.3.1: PMFs of X, Y and $S = X + Y$

1.2.4 Consider a frequency-modulated (FM) signal

$$f(t) = A_c \cos\left(2\pi f_c t + 3\sin\left(2\pi f_1 t\right) + 4\sin\left(6\pi f_1 t\right)\right) \tag{1.2.4.1}$$

where A_c and f_c are the amplitude and frequency (in Hz) of the carrier. The frequency f_1 is in Hz and $f_c > 100 f_1$. The peak frequency deviation of the FM signal in Hz is _____. (EC 2025)

a) $15 f_1$ b) $12 f_1$ c) $4 f_1$ d) $2 f_1$

Solution: The phase is

$$\phi(t) = 2\pi f_c t + 3\sin\left(2\pi f_1 t\right) + 4\sin\left(6\pi f_1 t\right) \tag{1.2.4.2}$$

From (**??**),

$$f_i(t) = \frac{1}{2\pi}\frac{d\phi}{dt} \tag{1.2.4.3}$$

$$= f_c + 3f_1 \cos\left(2\pi f_1 t\right) + 12 f_1 \cos\left(6\pi f_1 t\right) \tag{1.2.4.4}$$

$$\Delta f(t) = 3f_1 \cos\left(2\pi f_1 t\right) + 12 f_1 \cos\left(6\pi f_1 t\right) \tag{1.2.4.5}$$

Both cosines equal 1 at $t = 0$, so

$$\Delta f_{\text{peak}} = 3f_1 + 12f_1 \tag{1.2.4.6}$$
$$= 15f_1 \tag{1.2.4.7}$$

The following code plots the frequency deviation in Fig. **??**.

codes/FM_Peak_Deviation.py

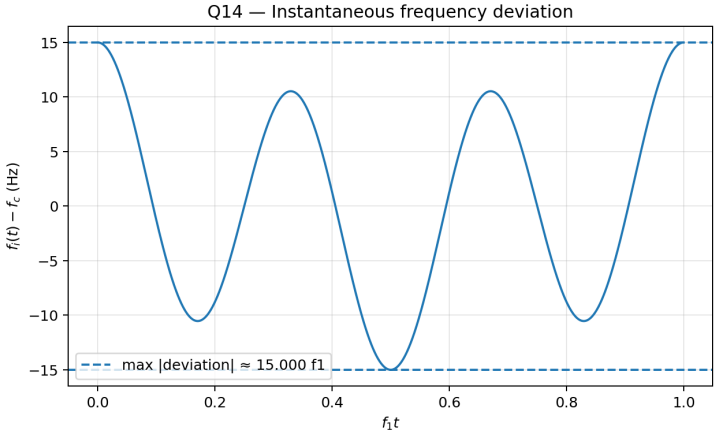


Fig. 1.2.4.1: Frequency deviation $\frac{\Delta f(t)}{f_1}$

1.2.5 Consider the discrete-time system in Fig. **??** with input $x[n]$ and output $y[n]$. Here $h_1[n]$ and $h_2[n]$ are the impulse responses of LTI Subsystems 1 and 2, $\delta[n]$ is the unit impulse, and $b > 0$. Assuming $h_2[n] \neq \delta[n]$, the overall system (dashed box) is _____. (EC 2025)

a) linear and time invariant
b) linear and time variant
c) nonlinear and time invariant
d) nonlinear and time variant

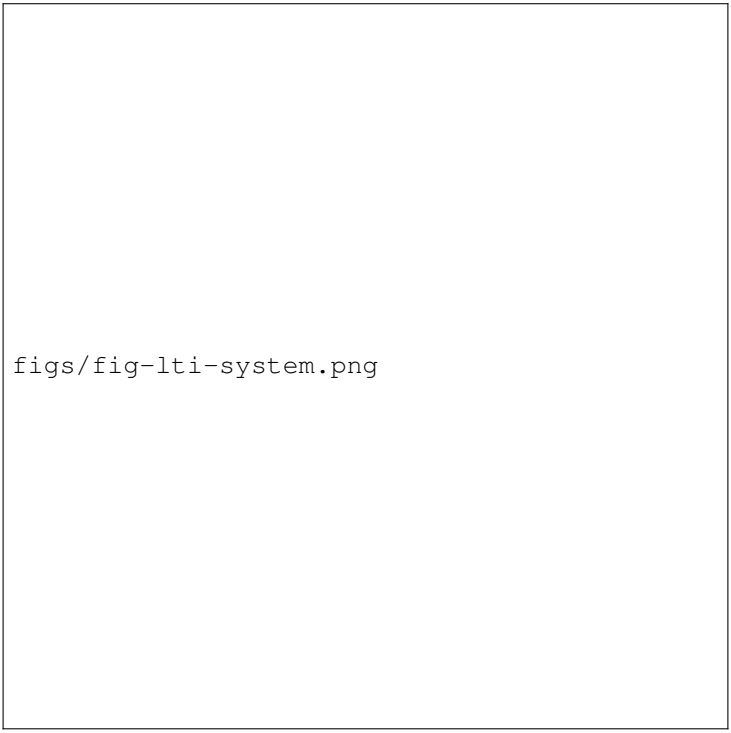


Fig. 1.2.5.1: Discrete-time system

Solution: From Fig. **??** and (**??**),

$$y[n] = h_2[n] * (h_1[n] * x[n] + b\delta[n]) - b\delta[n] \tag{1.2.5.1}$$
$$= (h_2 * h_1 * x)[n] + b(h_2[n] - \delta[n]) \tag{1.2.5.2}$$

For $x[n] = 0$,

$$y[n] = b(h_2[n] - \delta[n]) \neq 0 \tag{1.2.5.3}$$

which contradicts (**??**), so the system is nonlinear. For a delayed input,

$$T\{x[n-k]\} = (h_2 * h_1 * x)[n-k] + b(h_2[n] - \delta[n]) \tag{1.2.5.4}$$
$$y[n-k] = (h_2 * h_1 * x)[n-k] + b(h_2[n-k] - \delta[n-k]) \tag{1.2.5.5}$$

These differ, so (**??**) fails and the system is time variant. The following code simulates the result in Fig. **??**.

codes/LTI_System.py

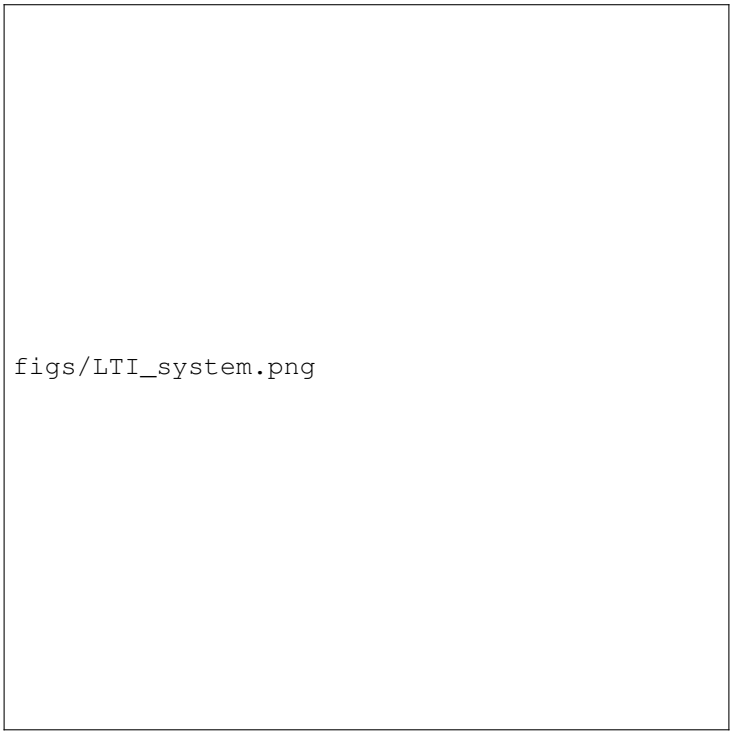


Fig. 1.2.5.2: Response to $x[n] = 0$ and to a shifted input

1.2.6 Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-\jmath\omega t} dt$ exists. Which one of the following statements is always TRUE? (EC 2025)

a) $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)| \, dt$
b) $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)| \, dt$
c) $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t) \, dt$
d) $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t) \, dt$

Solution: For $f(t) = e^{-|t|}$,

$$\int_{-\infty}^{\infty} |f(t)| \, dt = \int_{-\infty}^{\infty} f(t) \, dt = 2 \tag{1.2.6.1}$$

From (**??**),

$$|F(\omega)| = \frac{2}{1+\omega^2} \tag{1.2.6.2}$$

$$\leq 2 \tag{1.2.6.3}$$

with strict inequality for $\omega \neq 0$, so (B) and (D) fail. For $g(t) = -e^{-|t|}$,

$$|G(0)| = 2 \tag{1.2.6.4}$$

$$\int_{-\infty}^{\infty} g(t)\, dt = -2 \tag{1.2.6.5}$$

so (C) fails. From (**??**),

$$|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)| \left|e^{-\mathrm{J}\omega t}\right| dt \tag{1.2.6.6}$$

$$= \int_{-\infty}^{\infty} |f(t)|\, dt \tag{1.2.6.7}$$

for every signal, so (A) is true. The following code verifies the bound in Fig. **??**.

codes/Fourier_Bound.py

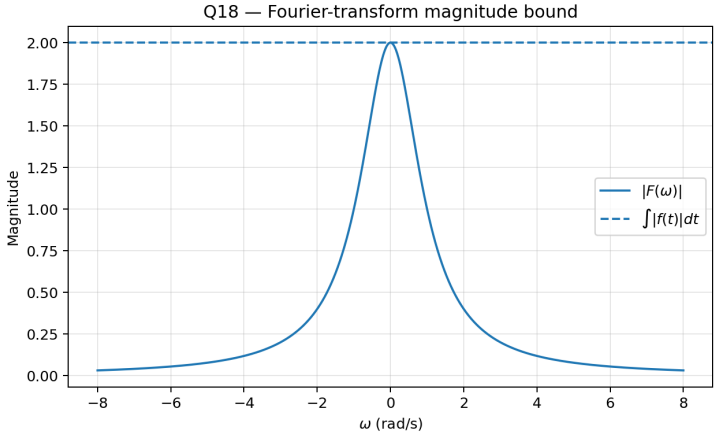


Fig. 1.2.6.1: $|F(\omega)|$ for $f(t) = e^{-|t|}$ against the bounds

1.2.7 Let i_C, i_L and i_R be the currents through the capacitor, inductor and resistor in the circuit of Fig. **??**. The AC admittances are given in Siemens (S). Which one of the following is TRUE? (EC 2025)

a) $i_C = 0.25\angle 180^\circ$ A, $i_L = 0.1\angle 0^\circ$ A, $i_R = 0.2\angle 90^\circ$ A

b) $i_C = 4\angle 180^\circ$ A, $i_L = 10\angle 0^\circ$ A, $i_R = 5\angle 90^\circ$ A

c) $i_C = 0.25\angle 270^\circ$ A, $i_L = 0.1\angle 90^\circ$ A, $i_R = 0.2\angle 90^\circ$ A

d) $i_C = 4\angle 90^\circ$ A, $i_L = 10\angle 270^\circ$ A, $i_R = 5\angle 0^\circ$ A

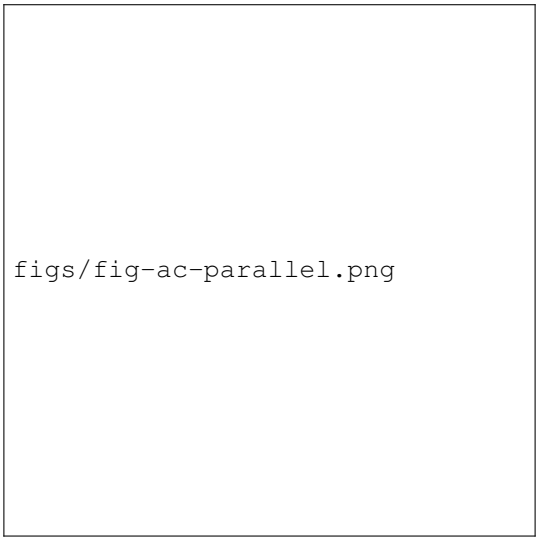


Fig. 1.2.7.1: Parallel R, L, C branches

Solution: From Fig. **??**, $\underline{V} = 1\angle 90^\circ$ V. From (**??**) and (**??**),

$$\underline{I}_C = (\jmath 0.25)\,(1\angle 90^\circ) \tag{1.2.7.1}$$
$$= 0.25\angle 180^\circ \text{ A} \tag{1.2.7.2}$$
$$\underline{I}_L = (-\jmath 0.1)\,(1\angle 90^\circ) \tag{1.2.7.3}$$
$$= 0.1\angle 0^\circ \text{ A} \tag{1.2.7.4}$$
$$\underline{I}_R = (0.2)\,(1\angle 90^\circ) \tag{1.2.7.5}$$
$$= 0.2\angle 90^\circ \text{ A} \tag{1.2.7.6}$$

The following code plots the phasors in Fig. **??**.

codes/AC_Admittance.py

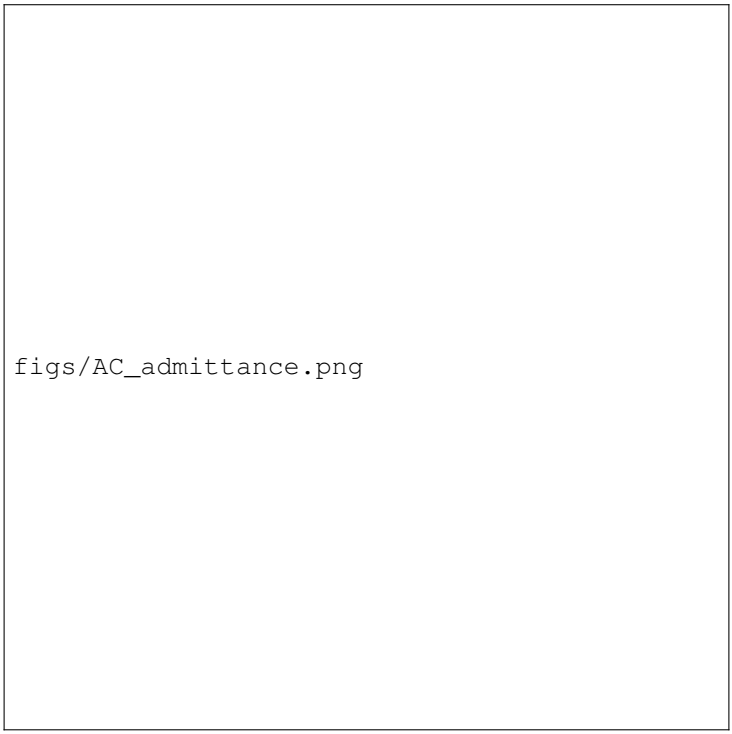


Fig. 1.2.7.2: Branch current phasors

1.2.8 A 3-input majority logic gate has inputs X, Y, Z. The output F is logic '1' if two or more inputs are logic '1', and logic '0' if two or more inputs are logic '0'. Which one of the following is a Boolean expression of F? (EC 2025)

a) $XY + YZ + ZX$b) $X \oplus Y \oplus Z$ c) $X + Y + Z$ d) XYZ

Solution: $F = 1$ for the inputs $011, 101, 110, 111$. Pairing each of the first three minterms with XYZ,

$$F = \overline{X}YZ + X\overline{Y}Z + XY\overline{Z} + XYZ \tag{1.2.8.1}$$
$$= \left(\overline{X}YZ + XYZ\right) + \left(X\overline{Y}Z + XYZ\right) + \left(XY\overline{Z} + XYZ\right) \tag{1.2.8.2}$$
$$= YZ + XZ + XY \tag{1.2.8.3}$$

which is (**??**). The truth table is

X	Y	Z	F	$XY+YZ+ZX$	$X\oplus Y\oplus Z$	$X+Y+Z$	XYZ
0	0	0	0	0	0	0	0
0	0	1	0	0	1	1	0
0	1	0	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	0	1	1	0
1	0	1	1	1	0	1	0
1	1	0	1	1	0	1	0
1	1	1	1	1	1	1	1

The following code verifies the truth table.

codes/Majority_Truth_Table.c

1.2.9 A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, Z and output F, as in Fig. **??**. The input Z is connected to the carry-in of the full adder. If Z is set to logic '1', the circuit functions as _____ with X and Y as inputs. (EC 2025)

a) an adder
b) a subtractor
c) a multiplier
d) a binary to Gray code converter

figs/fig-adder-xor.png

Fig. 1.2.9.1: Full adder with an XOR gate

Solution: From Fig. **??**, the XOR gate forms $Y\oplus Z$ and Z is the carry-in. For $Z=1$, from (**??**),

$$Y\oplus 1=\overline{Y} \tag{1.2.9.1}$$

From (**??**),

$$F=X+\overline{Y}+1 \tag{1.2.9.2}$$
$$\equiv X-Y \pmod{2^n} \tag{1.2.9.3}$$

so the circuit is a subtractor. The following code verifies the result in Fig. **??**.

codes/Full_Adder_Subtractor.py

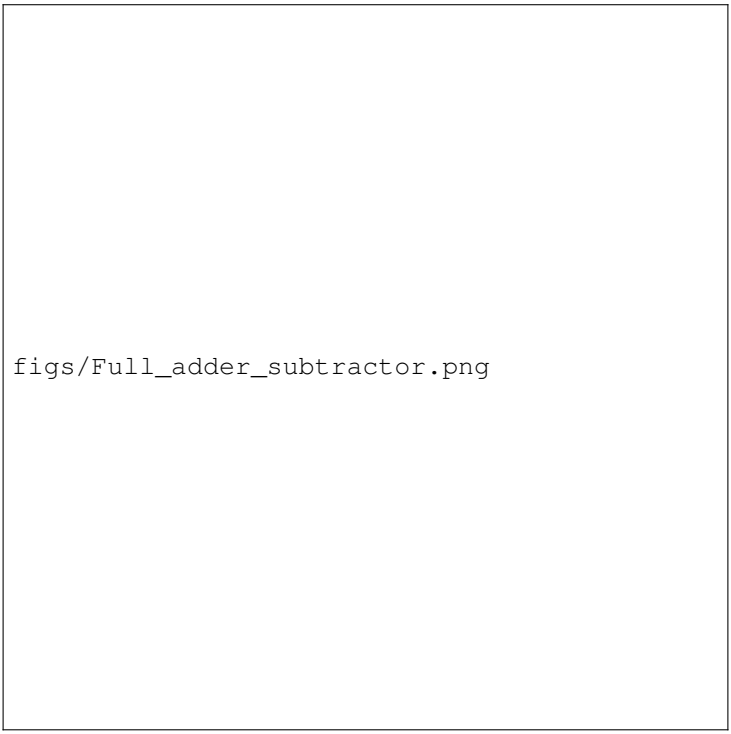


Fig. 1.2.9.2: Circuit output against $X - Y$

1.2.10 Consider $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x^3 - 3x^2 - 12x + 1$. Which of the following statements is/are correct? (EC 2025)

a) f has no global maximizer
b) f has no global minimizer
c) $x = -1$ is a local minimizer of f
d) $x = 2$ is a local maximizer of f

Solution:

$$f'(x) = 6x^2 - 6x - 12 \tag{1.2.10.1}$$
$$= 6(x+1)(x-2) \tag{1.2.10.2}$$
$$f''(x) = 12x - 6 \tag{1.2.10.3}$$

From (**??**),

$$f''(-1) = -18 < 0 \tag{1.2.10.4}$$
$$\implies x = -1 \text{ is a local maximum} \tag{1.2.10.5}$$
$$f''(2) = 18 > 0 \tag{1.2.10.6}$$
$$\implies x = 2 \text{ is a local minimum} \tag{1.2.10.7}$$

So (C) and (D) are false. Also,

$$\lim_{x\to\infty} f(x) = +\infty \tag{1.2.10.8}$$

$$\lim_{x\to-\infty} f(x) = -\infty \tag{1.2.10.9}$$

so f has neither a global maximizer nor a global minimizer. The following code plots the function in Fig. **??**.

codes/Cubic_Function.py

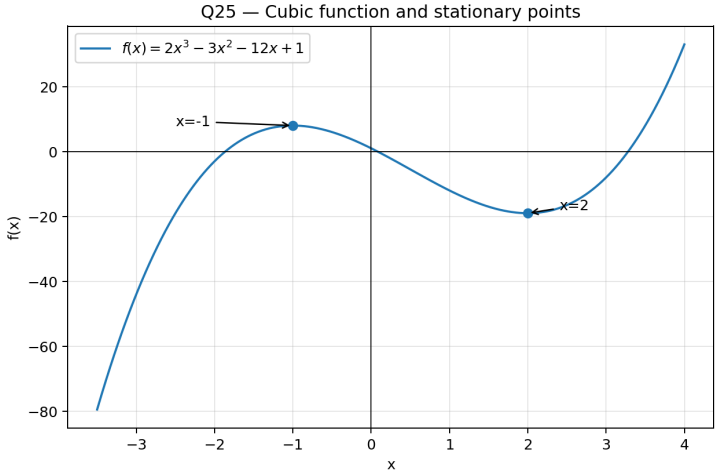


Fig. 1.2.10.1: $f(x) = 2x^3 - 3x^2 - 12x + 1$ with its local extrema

1.2.11 Consider the unity-negative-feedback system in Fig. **??**, where the gain $K \geq 0$. The root locus of this system is shown in Fig. **??**. For what value(s) of K will the system in Fig. **??** have a pole at $-1 + \jmath 1$? (EC 2025)

a) $K = 5$
b) $K = \frac{1}{5}$
c) For no positive value of K
d) For all positive values of K

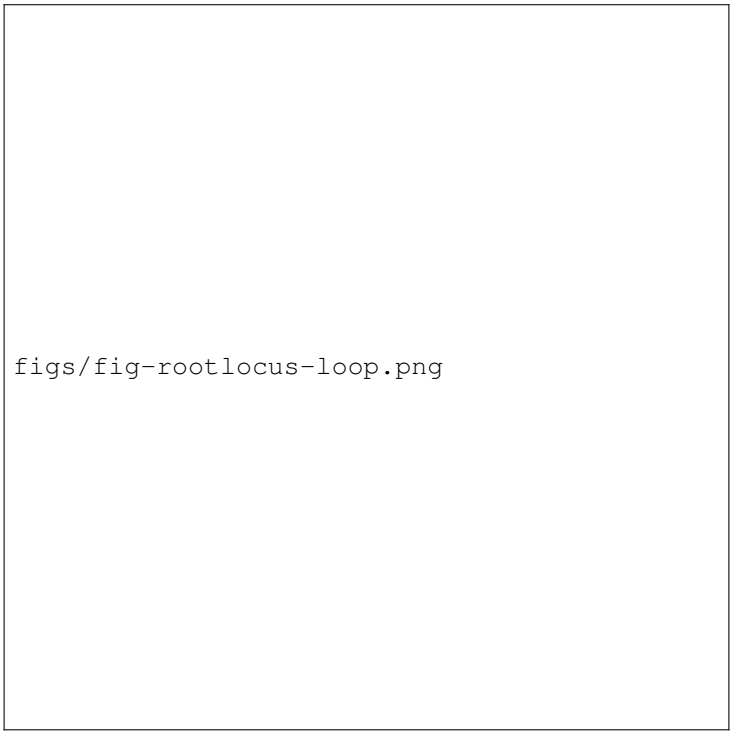


Fig. 1.2.11.1: Unity-negative-feedback system

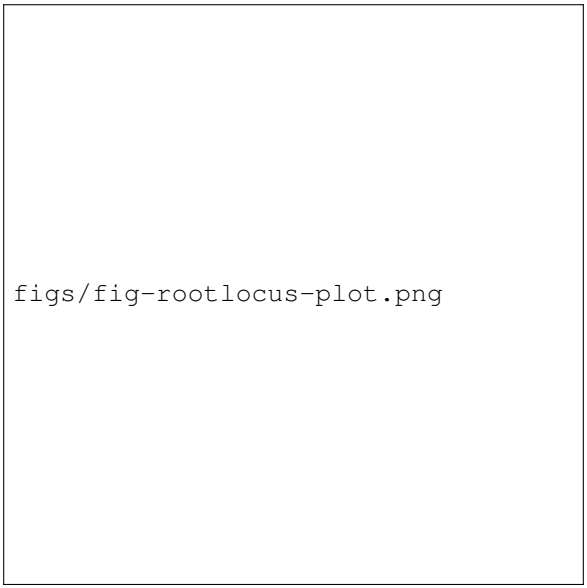


Fig. 1.2.11.2: Root locus of the system

Solution: From Fig. **??**,

$$G(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)} \tag{1.2.11.1}$$

From (**??**), at $s_0 = -1 + \mathrm{J}$,

$$G(s_0) = \frac{(2+\mathrm{J})(3+\mathrm{J})}{\mathrm{J}(1+\mathrm{J})} \tag{1.2.11.2}$$

$$= \frac{5+5\mathrm{J}}{-1+\mathrm{J}} \tag{1.2.11.3}$$

$$= -5\mathrm{J} \tag{1.2.11.4}$$

which is not real, so s_0 is not on the root locus for any K. The following code plots the root locus in Fig. **??**.

codes/Root_Locus.py

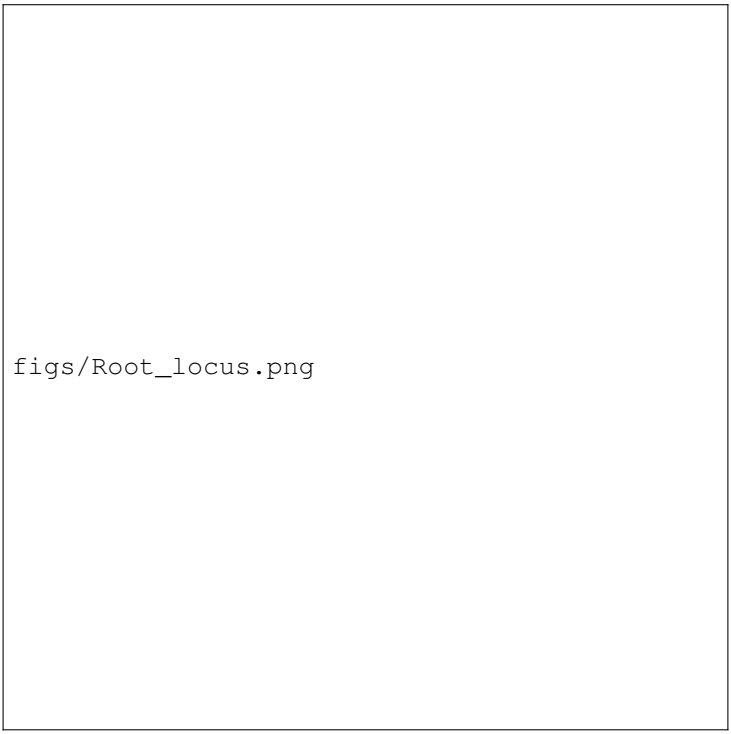


Fig. 1.2.11.3: Root locus with $s_0 = -1 + \jmath$ marked

1.2.12 Let $x[n]$ be a discrete-time signal whose z-transform is $X(z)$. Which of the following statements is/are TRUE? (EC 2025)

a) The discrete-time Fourier transform (DTFT) of $x[n]$ always exists
b) The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros
c) The DTFT exists if the RoC contains the unit circle
d) If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the RoC is the entire z-plane

Solution: For $x[n] = 2^n u[n]$ the RoC is $|z| > 2$, which excludes the unit circle, so (A) is false. For

$$x[n] = \delta[n] - 0.5\delta[n-1] \tag{1.2.12.1}$$

$$X(z) = 1 - 0.5z^{-1} \tag{1.2.12.2}$$

$$= \frac{z - 0.5}{z} \tag{1.2.12.3}$$

the zero $z = 0.5$ lies in the RoC, so (B) is false. From (**??**), (C) is true. For $x[n] =$

$\alpha\delta[n]$,

$$X(z) = \alpha \sum_{n} \delta[n] z^{-n} \tag{1.2.12.4}$$

$$= \alpha \tag{1.2.12.5}$$

which is finite for every z, so (D) is true. The following code verifies the result in Fig. **??**.

codes/Z_Transform_ROC.py

figs/Z_transform_ROC.png

Fig. 1.2.12.1: Partial sums for $2^n u[n]$ and $0.5^n u[n]$

1.2.13 Consider a message signal $m(t)$ bandlimited to $[-W, W]$ (W in Hz) and two schemes:

$$\text{DSB-SC: } f_{DSB}(t) = A_c m(t) \cos(2\pi f_c t) \tag{1.2.13.1}$$

$$\text{AM: } f_{AM}(t) = A_c (1 + \mu m(t)) \cos(2\pi f_c t) \tag{1.2.13.2}$$

Here A_c, f_c are the carrier amplitude and frequency, and μ is the modulation index. Consider the statements:
(i) An envelope detector can be used for demodulation in the DSB-SC scheme if

$m(t) > 0$ for all t.
(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.
Which of the following options is/are correct? (EC 2025)

a) (i) is TRUE b) (i) is FALSE c) (ii) is TRUE d) (ii) is FALSE

Solution: From (**??**), for DSB-SC with $m(t) > 0$,

$$A_c |m(t)| = A_c m(t) \tag{1.2.13.3}$$

so (i) is true. For AM, recovery needs only $1 + \mu m(t) \geq 0$. For $m(t) = \cos(2\pi W t)$ and $\mu = \frac{1}{2}$,

$$1 + \mu m(t) \geq \frac{1}{2} \tag{1.2.13.4}$$

$$> 0 \tag{1.2.13.5}$$

although $m(t)$ is negative half the time, so (ii) is false. The following code plots the envelopes in Fig. **??**.

codes/Envelope_Detection.py

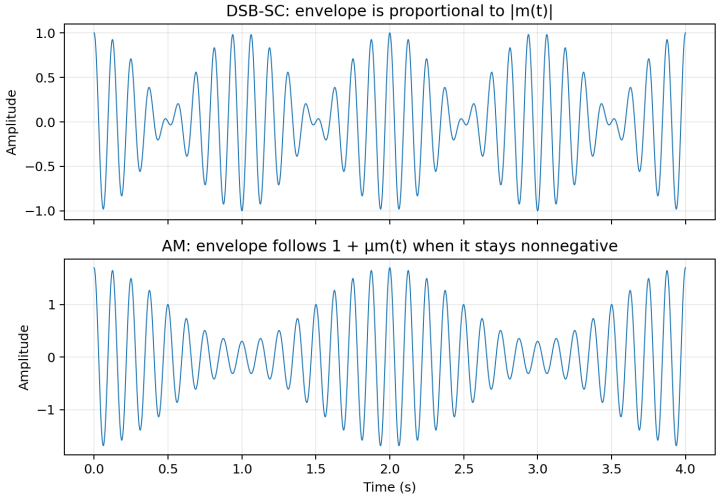


Fig. 1.2.13.1: Envelopes of the DSB-SC and AM waves

1.2.14 The function $y(t)$ satisfies $t^2 y''(t) - 2ty'(t) + 2y(t) = 0$, where y' and y'' are the first and second derivatives. Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value

of $y(t)$ over $[0,1]$ is _____ (rounded off to two decimal places). (EC 2025)

Solution: From (**??**),

$$m(m-1)-2m+2=0 \tag{1.2.14.1}$$
$$\implies (m-1)(m-2)=0 \tag{1.2.14.2}$$
$$\implies y=C_1t+C_2t^2 \tag{1.2.14.3}$$
$$y'=C_1+2C_2t \tag{1.2.14.4}$$

Substituting $y'(0)=1$ and $y'(1)=-1$,

$$C_1=1 \tag{1.2.14.5}$$
$$1+2C_2=-1 \tag{1.2.14.6}$$
$$\implies C_2=-1 \tag{1.2.14.7}$$
$$y=t-t^2 \tag{1.2.14.8}$$

From (**??**),

$$y'=1-2t=0 \tag{1.2.14.9}$$
$$\implies t=\frac{1}{2} \tag{1.2.14.10}$$
$$y_{\max}=\frac{1}{2}-\frac{1}{4} \tag{1.2.14.11}$$
$$=0.25 \tag{1.2.14.12}$$

1.2.15 The generator matrix of a $(6,3)$ binary linear block code is

$$\mathbf{G}=\begin{pmatrix}1&0&0&1&0&1\\0&1&0&0&1&1\\0&0&1&1&1&0\end{pmatrix} \tag{1.2.15.1}$$

The minimum Hamming distance $d_{\min}$ between codewords equals _____ (answer in integer). (EC 2025)

Solution: The codewords are $\mathbf{c}=\mathbf{uG}$ (mod 2):

$\mathbf{u}$	$\mathbf{c}$	$w(\mathbf{c})$
001	001110	3
010	010011	3
011	011101	4
100	100101	3
101	101011	4
110	110110	4
111	111000	3

From (**??**),

$$d_{\min}=\min_{\mathbf{c}\neq\mathbf{0}} w(\mathbf{c}) \tag{1.2.15.2}$$
$$=3 \tag{1.2.15.3}$$

The following code verifies the solution in Fig. **??**.

codes/Minimum_Hamming_Distance.py

figs/Minimum_hamming_distance.png

Fig. 1.2.15.1: Weights of the non-zero codewords

1.2.16 All components in the bandpass filter of Fig. **??** are ideal. The lower -3 dB frequency is 1 MHz. The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____. (EC 2025)



Fig. 1.2.16.1: Active bandpass filter

Solution: From Fig. **??**,

$$Z_i = R + \frac{1}{s10C} \tag{1.2.16.1}$$

$$= \frac{1 + 10sRC}{10sC} \tag{1.2.16.2}$$

$$Z_f = \frac{2R}{1 + 0.2sRC} \tag{1.2.16.3}$$

From (**??**),

$$H(s) = -\frac{Z_f}{Z_i} \tag{1.2.16.4}$$

$$= -\frac{20sRC}{(1 + 10sRC)(1 + 0.2sRC)} \tag{1.2.16.5}$$

From (**??**),

$$\omega_L = \frac{1}{10RC} \tag{1.2.16.6}$$

$$\omega_H = \frac{1}{0.2RC} \tag{1.2.16.7}$$

$$\frac{f_H}{f_L} = \frac{10RC}{0.2RC} \tag{1.2.16.8}$$

$$= 50 \tag{1.2.16.9}$$

$$f_H = 50 \times 1 \text{ MHz} \tag{1.2.16.10}$$

$$= 50 \text{ MHz} \tag{1.2.16.11}$$

The following code plots the magnitude response in Fig. **??**.

codes/Bandpass_Cutoff.py

figs/Bandpass_cutoff.png

Fig. 1.2.16.2: Magnitude response of the bandpass filter

1.2.17 Let $G(s) = \frac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to $G(s)$ in the configuration of Fig. **??**. Consider:
(i) There exists no controller $M(s) = \frac{K_I}{s}$, with K_I a positive real number, such that

the closed-loop system is stable.
(ii) There exists at least one controller $M(s) = K_P + sK_D$, with K_P, K_D positive real numbers, such that the closed-loop system is stable.
Which one of the following options is correct? (EC 2025)

a) (i) is TRUE and (ii) is FALSE
b) (i) is FALSE and (ii) is TRUE
c) Both (i) and (ii) are FALSE
d) Both (i) and (ii) are TRUE

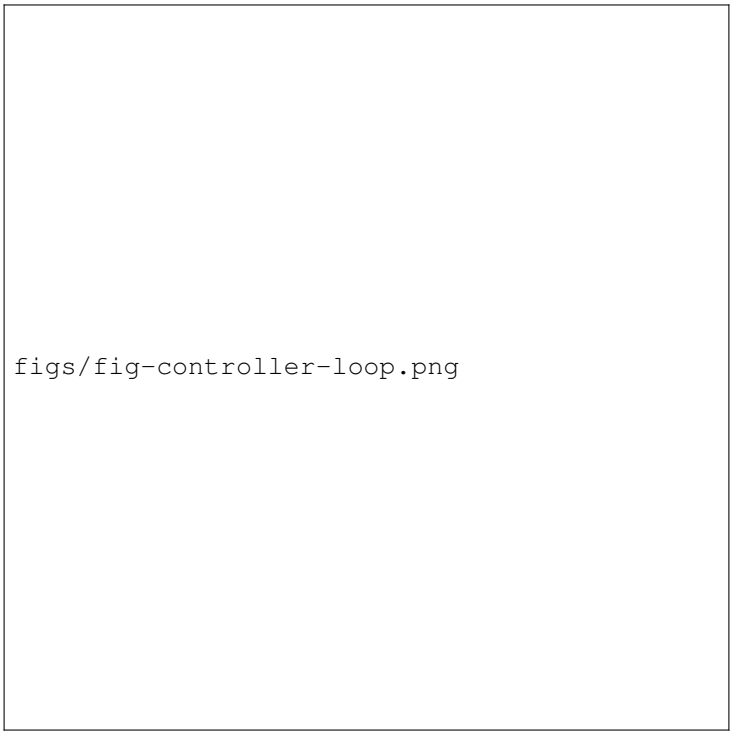


Fig. 1.2.17.1: Controller $M(s)$ and plant $G(s)$ in feedback

Solution: From (**??**), for $M(s) = \frac{K_I}{s}$,

$$1 + \frac{K_I}{s} \cdot \frac{1}{10s^2} = 0 \tag{1.2.17.1}$$

$$\implies 10s^3 + K_I = 0 \tag{1.2.17.2}$$

From (**??**),

$$s = \left(\frac{K_I}{10}\right)^{\frac{1}{3}} e^{\mathrm{J}\frac{\pi+2\pi k}{3}} \tag{1.2.17.3}$$

The roots are at $60^\circ, 180^\circ, 300^\circ$, so two lie in the right half-plane and (i) is true. For $M(s) = K_P + sK_D$,

$$1 + \frac{K_P + sK_D}{10s^2} = 0 \tag{1.2.17.4}$$

$$\implies 10s^2 + K_D s + K_P = 0 \tag{1.2.17.5}$$

All coefficients are positive, so from (**??**), (ii) is true. The following code plots the closed-loop poles in Fig. **??**.

codes/Controller_Stability.py

figs/Controller_stability.png

Fig. 1.2.17.2: Closed-loop poles for the integral and PD controllers

1.2.18 Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as: L is the number of roots of $p(s)$ with negative real parts, I is the number of purely imaginary roots, and R is the number of roots with positive real parts. Which one of the following is correct? (EC 2025)

a) $L=2$, $I=2$, and $R=1$
b) $L=3$, $I=2$, and $R=0$
c) $L=1$, $I=2$, and $R=2$
d) $L=0$, $I=4$, and $R=1$

Solution:

$$(s+5)(s+4)=s^2+9s+20 \tag{1.2.18.1}$$
$$\left(s^2+9s+20\right)(s-2)=s^3+7s^2+2s-40 \tag{1.2.18.2}$$
$$\left(s^3+7s^2+2s-40\right)\left(s^2+1\right)=s^5+7s^4+3s^3-33s^2+2s-40 \tag{1.2.18.3}$$
$$\implies p(s)=(s+5)(s+4)(s-2)\left(s^2+1\right) \tag{1.2.18.4}$$

The roots $-5,-4$ have negative real parts, $\pm\mathrm{J}$ are purely imaginary and 2 has a positive real part, so

$$(L,I,R)=(2,2,1) \tag{1.2.18.5}$$

The following code plots the roots in Fig. **??**.

codes/Routh_Hurwitz.py

figs/Routh_hurwitz.png

Fig. 1.2.18.1: Roots of $p(s)$

1.2.19 Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside $[-\omega_c, \omega_c]$ (ω_c in rad/s). It is uniformly sampled to obtain $y(t) = f(t)\,p(t)$, where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s), \quad T_s > 0,\ \tau > 0 \tag{1.2.19.1}$$

and $\delta(t)$ is the Dirac impulse. The sampled signal $y(t)$ is passed through an ideal lowpass filter

$$h(t) = \omega_c T_s \frac{\sin(\omega_c t)}{\pi \omega_c t} \tag{1.2.19.2}$$

with cutoff frequency ω_c and passband gain T_s. The output of the filter is _____. (EC 2025)

a) $f(t)$ if $T_s < \frac{\pi}{\omega_c}$
b) $f(t-\tau)$ if $T_s < \frac{\pi}{\omega_c}$
c) $f(t-\tau)$ if $T_s < \frac{2\pi}{\omega_c}$
d) $T_s f(t)$ if $T_s < \frac{2\pi}{\omega_c}$

Solution: From (**??**),

$$Y(\omega) = \frac{1}{T_s} \sum_{k} e^{-\mathrm{j}k\omega_s \tau} F(\omega - k\omega_s) \tag{1.2.19.3}$$

From (**??**), the replicas do not overlap if

$$\omega_s = \frac{2\pi}{T_s} > 2\omega_c \tag{1.2.19.4}$$

$$\iff T_s < \frac{\pi}{\omega_c} \tag{1.2.19.5}$$

From (**??**), only $k = 0$ is kept,

$$H(\omega)\,Y(\omega) = T_s \cdot \frac{1}{T_s} F(\omega) \tag{1.2.19.6}$$

$$= F(\omega) \tag{1.2.19.7}$$

so the output is $f(t)$. The following code verifies the reconstruction in Fig. **??**.

codes/Impulse_Sampling.py

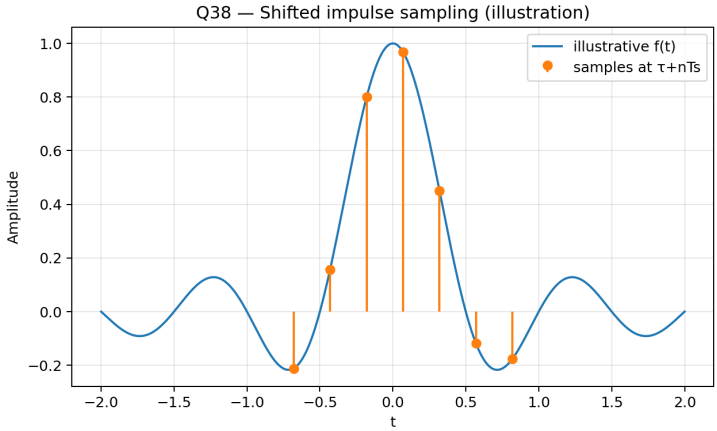


Fig. 1.2.19.1: Reconstruction equals $f\,(t)$

1.2.20 In the circuit of Fig. **??**, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s\,(t) = 100\cos\,(200t)$. What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively? (EC 2025)

a) 25 μF
b) 4 μF
c) 0.25 μF
d) Insufficient information to find C

figs/fig-ac-meters.png

Fig. 1.2.20.1: Circuit with meters M_1, M_2 and capacitor C

Solution: From Fig. **??**,

$$|Z| = \frac{V}{I} \tag{1.2.20.1}$$
$$= \frac{25}{5} \tag{1.2.20.2}$$
$$= 5\ \Omega \tag{1.2.20.3}$$

From (**??**),

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = R \tag{1.2.20.4}$$
$$\implies \omega L = \frac{1}{\omega C} \tag{1.2.20.5}$$

From (**??**),

$$C = \frac{1}{\omega^2 L} \tag{1.2.20.6}$$
$$= \frac{1}{200^2 \times 1} \tag{1.2.20.7}$$
$$= 25\ \mu\text{F} \tag{1.2.20.8}$$

The following code verifies the solution in Fig. **??**.

codes/RLC_Capacitor.py

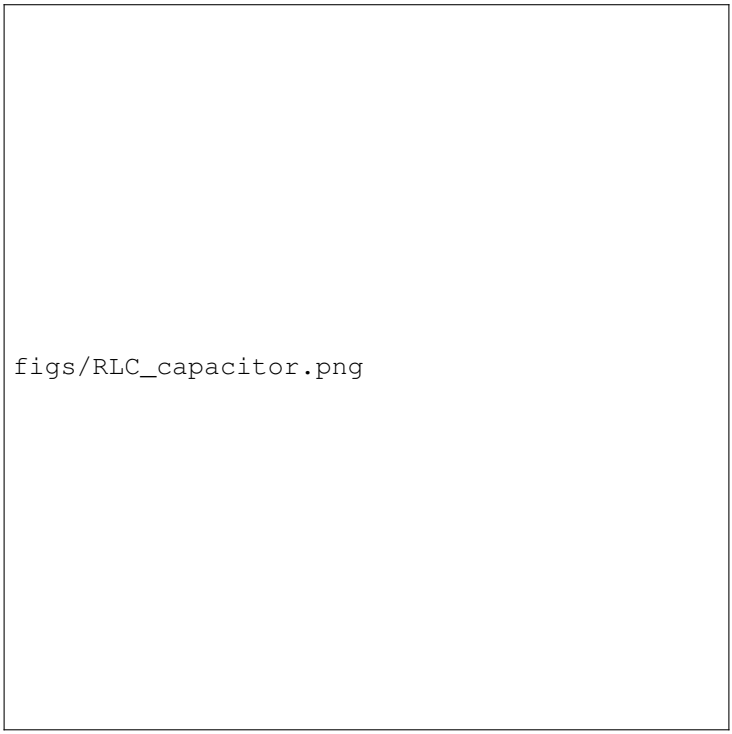


Fig. 1.2.20.2: $|Z|$ against C

1.2.21 The Z-parameter matrix of a two-port network relates port voltages and currents by

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \mathbf{Z} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \tag{1.2.21.1}$$

The Z-parameter matrix (each entry in Ohms) of the network in Fig. **??** is _____. (EC 2025)

a) $\begin{pmatrix} \frac{10}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{10}{3} \end{pmatrix}$ b) $\begin{pmatrix} \frac{2}{3} & \frac{10}{3} \\ \frac{10}{3} & \frac{2}{3} \end{pmatrix}$ c) $\begin{pmatrix} 10 & 2 \\ 2 & 10 \end{pmatrix}$ d) $\begin{pmatrix} \frac{10}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{10}{3} \end{pmatrix}$

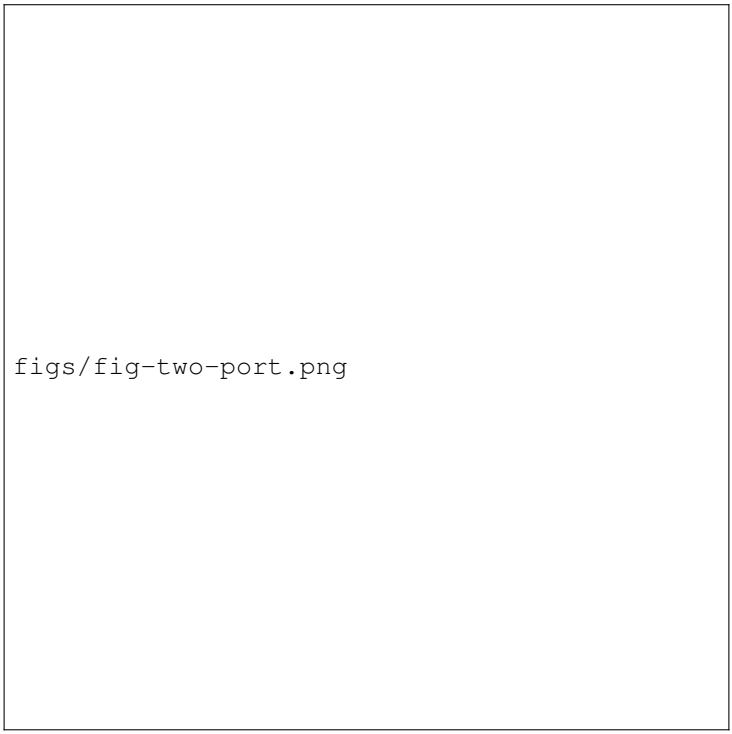


Fig. 1.2.21.1: Resistive two-port network

Solution: With port 2 open, from (**??**),

$$R_A = 2 \parallel (2+2) \tag{1.2.21.2}$$
$$= \frac{2 \cdot 4}{2+4} \tag{1.2.21.3}$$
$$= \frac{4}{3}\ \Omega \tag{1.2.21.4}$$

From (**??**),

$$z_{11} = 2 + R_A \tag{1.2.21.5}$$
$$= \frac{10}{3}\ \Omega \tag{1.2.21.6}$$
$$V_2 = \frac{2}{2+2} I_1 R_A \tag{1.2.21.7}$$
$$= \frac{2}{3} I_1 \tag{1.2.21.8}$$
$$z_{21} = \frac{2}{3}\ \Omega \tag{1.2.21.9}$$

The network is reciprocal and symmetric, so from (**??**),

$$\mathbf{Z} = \begin{pmatrix} \frac{10}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{10}{3} \end{pmatrix} \ \Omega \tag{1.2.21.10}$$

1.2.22 A source transmits a symbol S taking values uniformly at random from $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S, and uses the maximum-likelihood decoder. Suppose the probability of symbol error is $P_e = \alpha \Pr(N > 1)$. What is α? (EC 2025)

a) $\frac{1}{3}$ b) 1 c) $\frac{2}{3}$ d) $\frac{4}{3}$

Solution: From (**??**), the decision boundaries are -1 and 1. Let $p = \Pr(N > 1)$. From (**??**),

$$\Pr(\text{error}|S = -2) = \Pr(N > 1) = p \tag{1.2.22.1}$$

$$\Pr(\text{error}|S = 0) = \Pr(|N| > 1) = 2p \tag{1.2.22.2}$$

$$\Pr(\text{error}|S = 2) = \Pr(N < -1) = p \tag{1.2.22.3}$$

$$P_e = \frac{1}{3}(p + 2p + p) \tag{1.2.22.4}$$

$$= \frac{4}{3}p \tag{1.2.22.5}$$

so $\alpha = \frac{4}{3}$. The following code simulates the result in Fig. **??**.

codes/ML_Error_Probability.py

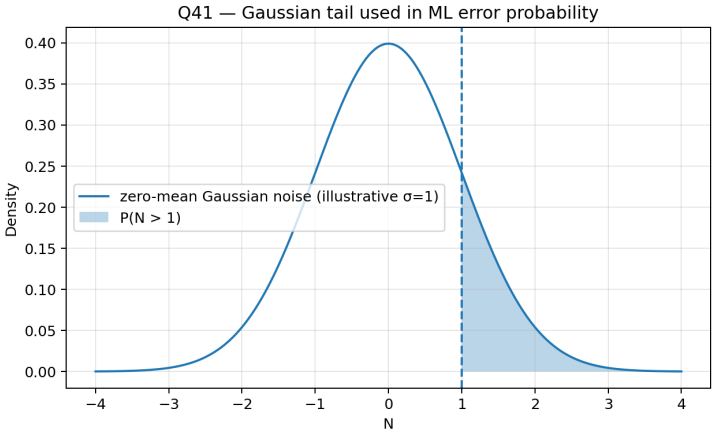


Fig. 1.2.22.1: Likelihoods of the three symbols and the ML thresholds

1.2.23 Consider the real-valued random process

$$f(t) = \sum_{n=1}^{N} a_n p(t - nT) \tag{1.2.23.1}$$

where $T > 0$, N is a positive integer, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise, and the coefficients a_n are pairwise independent, zero-mean, unit-variance random variables. Consider:
(i) The mean of the process $f(t)$ is independent of time t.
(ii) The autocorrelation $E[f(t) f(t+\tau)]$ is independent of t for all τ.
Choose the correct option. (EC 2025)

a) (i) is TRUE and (ii) is FALSE
b) Both (i) and (ii) are TRUE
c) Both (i) and (ii) are FALSE
d) (i) is FALSE and (ii) is TRUE

Solution: From (**??**),

$$E[f(t)] = \sum_{n=1}^{N} E[a_n] p(t - nT) \tag{1.2.23.2}$$

$$= 0 \tag{1.2.23.3}$$

so (i) is true. From (**??**),

$$E[f(t) f(t+\tau)] = \sum_{n=1}^{N} p(t - nT) p(t + \tau - nT) \tag{1.2.23.4}$$

For $\tau = 0$,

$$E\left[f(t)^2\right] = \sum_{n=1}^{N} p(t - nT) \tag{1.2.23.5}$$

$$\in \{0, 1\} \tag{1.2.23.6}$$

which depends on t, so (ii) is false. The following code simulates the process in Fig. **??**.

codes/Random_Process.py

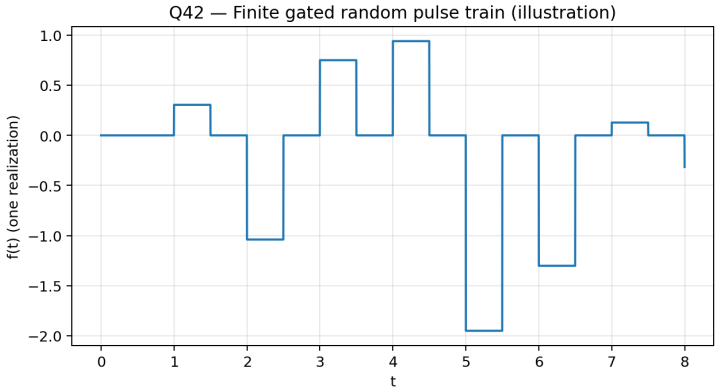


Fig. 1.2.23.1: $E\left[f\left(t\right)\right]=0$ but $E\left[f\left(t\right)^{2}\right]$ depends on t

1.2.24 A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full-scale voltage of 3.3 V. For an input sinusoid of frequency 500 kHz, the maximum SNR (in dB, two decimal places) and the data rate (in Mbps) at the ADC output are _____, respectively. (EC 2025)

a) 61.96 and 10 b) 61.96 and 5 c) 33.36 and 10 d) 33.36 and 5

Solution: From (**??**), for $N=10$ and $f_s=1$ MHz,

$$\begin{aligned}\text{SNR}_{\max} &= 6.02\left(10\right)+1.76 && (1.2.24.1)\\ &= 61.96 \text{ dB} && (1.2.24.2)\\ R_b &= f_s N && (1.2.24.3)\\ &= \left(10^{6}\right)\left(10\right) && (1.2.24.4)\\ &= 10 \text{ Mbps} && (1.2.24.5)\end{aligned}$$

The following code plots the SNR in Fig. **??**.

codes/ADC_SNR_Data_Rate.py



Fig. 1.2.24.1: Maximum SNR of an ideal ADC against the number of bits

1.2.25 A positive-edge-triggered sequential circuit is shown in Fig. **??**. There are no timing violations. Input P_0 is logic '0' and P_1 is logic '1' at all times. The timing diagram of SEL and S is also shown. The sequence of output Y from time T_0 to T_3 is _____. (EC 2025)

a) 1011 b) 0100 c) 0010 d) 1101

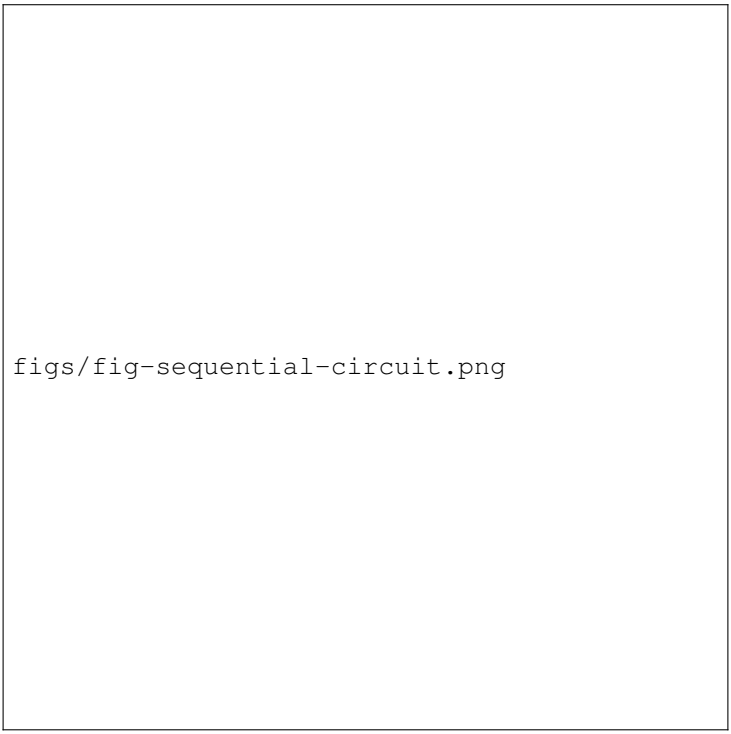


Fig. 1.2.25.1: Sequential circuit and timing diagram of SEL and S

Solution: From Fig. **??** and (**??**),

$$Q_1^+ = \begin{cases} S, & SEL = 1 \\ 0, & SEL = 0 \end{cases} \tag{1.2.25.1}$$

$$Q_2^+ = \begin{cases} Q_1, & SEL = 1 \\ 1, & SEL = 0 \end{cases} \tag{1.2.25.2}$$

$$Y = \overline{Q_1} \oplus \overline{Q_2} \tag{1.2.25.3}$$

$$= Q_1 \oplus Q_2 \tag{1.2.25.4}$$

Reading SEL and S at the rising edges before $T_0, \ldots, T_3$,

edge before	SEL	S	Q_1	Q_2	Y
T_0	0	0	0	1	1
T_1	1	0	0	0	0
T_2	1	1	1	0	1
T_3	0	1	0	1	1

The output sequence is $Y = 1011$. The following code verifies the solution.

codes/Sequential_Circuit_Verification.c

1.2.26 A square metal sheet of 4 m $\times$ 4 m is placed on the x–y plane as in Fig. **??**. If the surface charge density (in μC/m^2) is $\rho_s\,(x, y) = 4\,|y|$, the total charge (in μC, rounded off to the nearest integer) on the sheet is _____. (EC 2025)

a) 16 b) 85 c) 64 d) 256

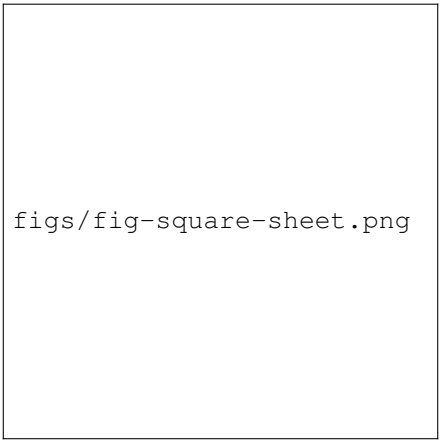


Fig. 1.2.26.1: Square metal sheet in the x–y plane

Solution: From Fig. **??**, the sheet is $-2 \le x \le 2$, $-2 \le y \le 2$. From (**??**),

$$Q = \int_{-2}^{2}\int_{-2}^{2} 4\,|y|\;dx\,dy \tag{1.2.26.1}$$

$$= 4\left(\int_{-2}^{2} dx\right)\left(2\int_{0}^{2} y\,dy\right) \tag{1.2.26.2}$$

$$= 4 \cdot 4 \cdot 4 \tag{1.2.26.3}$$

$$= 64\ \mu\text{C} \tag{1.2.26.4}$$

The following code plots the charge density in Fig. **??**.

codes/Surface_Charge.py

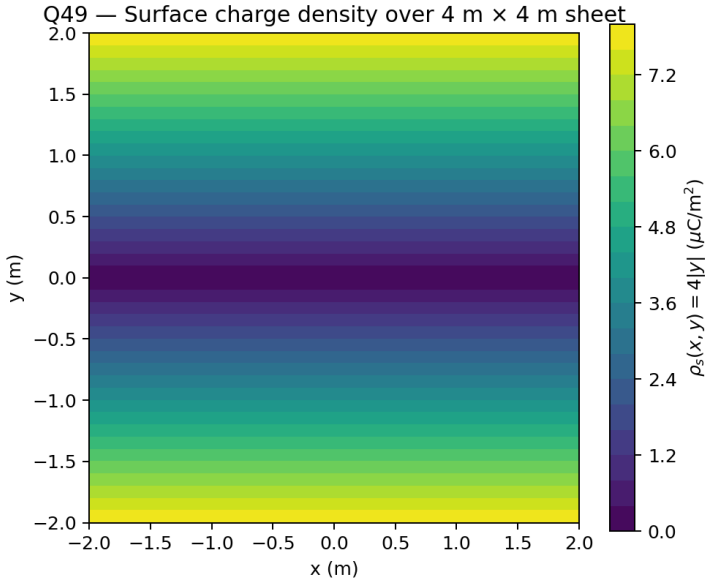


Fig. 1.2.26.2: Charge density $\rho_s = 4\,|y|$ on the sheet

1.2.27 An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has conductivity 5.8×10^7 S/m. The current (in Amperes, two decimal places) through the wire is _____. (EC 2025)

a) 0.46 b) 1.82 c) 0.58 d) 1.12

Solution: From (**??**), with $r = 0.5$ mm,

$$J = \sigma E \tag{1.2.27.1}$$
$$= \left(5.8 \times 10^7\right)(0.01) \tag{1.2.27.2}$$
$$= 5.8 \times 10^5 \text{ A/m}^2 \tag{1.2.27.3}$$
$$I = JA \tag{1.2.27.4}$$
$$= 5.8 \times 10^5 \cdot \pi \left(5 \times 10^{-4}\right)^2 \tag{1.2.27.5}$$
$$= 0.4555 \text{ A} \tag{1.2.27.6}$$
$$\approx 0.46 \text{ A} \tag{1.2.27.7}$$

1.2.28 Consider a non-negative function $f(x)$, continuous and bounded on $[2, 8]$. Let M and m be its maximum and minimum on the interval. Choose the combination(s) of α and β for which

$$\beta \leq \int_2^8 f(x)\, dx \leq \alpha \tag{1.2.28.1}$$

is guaranteed to hold. (EC 2025)

a) $\beta = 5m,\ \alpha = 7M$
b) $\beta = 6m,\ \alpha = 5M$
c) $\beta = 7m,\ \alpha = 6M$
d) $\beta = 7m,\ \alpha = 5M$

Solution: From (**??**), with $b - a = 6$,

$$6m \le \int_2^8 f(x)\, dx \le 6M \tag{1.2.28.2}$$

The inequality is guaranteed when $\beta \le 6m$ and $\alpha \ge 6M$. Only (A) satisfies

$$5m \le 6m \tag{1.2.28.3}$$

$$7M \ge 6M \tag{1.2.28.4}$$

The following code verifies the bounds in Fig. **??**.

codes/Integral_Bounds.py

figs/Integral_bounds.png

Fig. 1.2.28.1: The integral lies between $6m$ and $6M$

1.2.29 Which of the following statements about contour integrals (counter-clockwise) on the unit circle C in the complex plane is/are TRUE? (EC 2025)

a) $\oint_C e^z\,dz = 0$
b) $\oint_C z^n\,dz = 0$, where n is an even integer
c) $\oint_C \cos z\,dz \neq 0$
d) $\oint_C \sec z\,dz \neq 0$

Solution: e^z and $\cos z$ are analytic everywhere. From (**??**),

$$\oint_C e^z\,dz = 0 \tag{1.2.29.1}$$

$$\oint_C \cos z\,dz = 0 \tag{1.2.29.2}$$

so (A) is true and (C) is false. From (**??**), for even n,

$$\oint_C z^n\,dz = 0 \tag{1.2.29.3}$$

so (B) is true. From (**??**), the poles of $\sec z$ lie outside $|z| = 1$, so from (**??**),

$$\oint_C \sec z\,dz = 0 \tag{1.2.29.4}$$

and (D) is false. The following code computes the integrals in Fig. **??**.

codes/Contour_Integrals.py

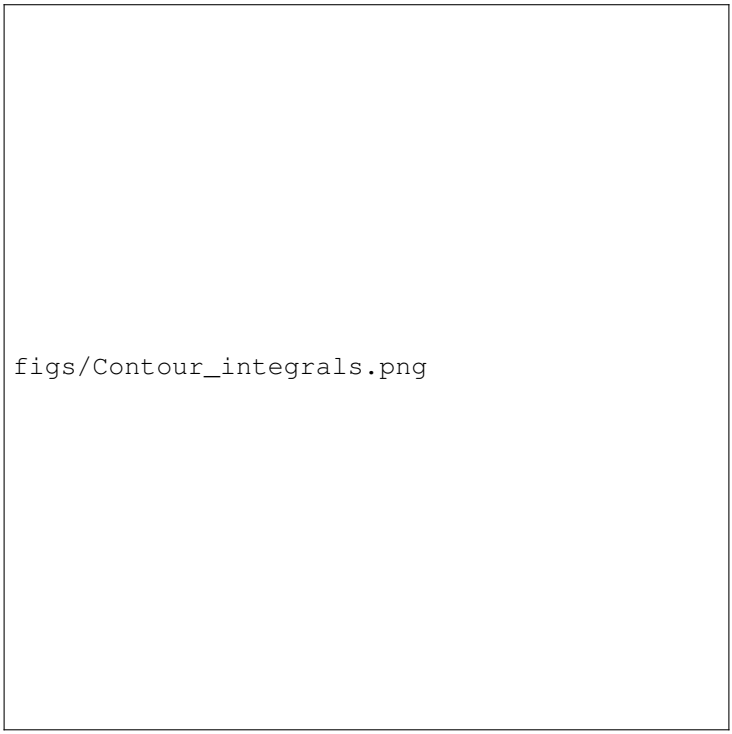


Fig. 1.2.29.1: Unit circle and the poles of $\sec z$ and $\frac{1}{z}$

1.2.30 Consider a system with internal states $x_1(t), x_2(t), x_3(t)$ and input $u(t)$ governed by

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} u(t) \tag{1.2.30.1}$$

Which of the following statements is/are TRUE? (EC 2025)

a) The signals x_1, x_2, x_3 are bounded for all bounded inputs.
b) There exists a bounded input such that at least one of x_1, x_2, x_3 is unbounded.
c) There exists a bounded input such that the signals x_1, x_2, x_3 are unbounded.
d) The signals x_1, x_2, x_3 are unbounded for all bounded inputs.

Solution: The states decouple,

$$\dot{x}_1 = 2x_1 + u \tag{1.2.30.2}$$
$$\dot{x}_2 = -2x_2 + u \tag{1.2.30.3}$$
$$\dot{x}_3 = u \tag{1.2.30.4}$$

For $u \equiv 1$ and zero initial state, from (**??**), (**??**) and (**??**),

$$x_1(t) = \frac{1}{2}\left(e^{2t} - 1\right) \to \infty \tag{1.2.30.5}$$

$$x_3(t) = t \to \infty \tag{1.2.30.6}$$

$$|x_2(t)| \leq |x_2(0)| + \frac{U}{2} \tag{1.2.30.7}$$

So (B) is true, while (A), (C) and (D) are false. The following code plots the state responses in Fig. **??**.

codes/State_Boundedness.py

figs/State_boundedness.png

Fig. 1.2.30.1: State responses to $u \equiv 1$ with zero initial state

1.2.31 The random variable X takes values in $\{-1, 0, 1\}$ with $\Pr(X = -1) = \Pr(X = 1) = \alpha$ and $\Pr(X = 0) = 1 - 2\alpha$, where $0 < \alpha < \frac{1}{2}$. Let $g(\alpha)$ be the entropy of X in bits. Which of the following is/are TRUE? (EC 2025)

a) $g\left(0.4\right) > g\left(0.3\right)$
b) $g\left(0.3\right) > g\left(0.4\right)$
c) $g\left(0.3\right) > g\left(0.25\right)$
d) $g\left(0.25\right) > g\left(0.3\right)$

Solution: From (**??**),

$$g\left(\alpha\right) = -2\alpha\log_2\alpha - \left(1-2\alpha\right)\log_2\left(1-2\alpha\right) \tag{1.2.31.1}$$

$$g'\left(\alpha\right) = 2\log_2\frac{1-2\alpha}{\alpha} \tag{1.2.31.2}$$

$g' > 0$ for $\alpha < \frac{1}{3}$. Evaluating,

$$g\left(0.25\right) = 1.5000 \tag{1.2.31.3}$$

$$g\left(0.3\right) = 1.5710 \tag{1.2.31.4}$$

$$g\left(0.4\right) = 1.5219 \tag{1.2.31.5}$$

Hence (B) and (C) are true. The following code plots the entropy in Fig. **??**.

codes/Entropy.py

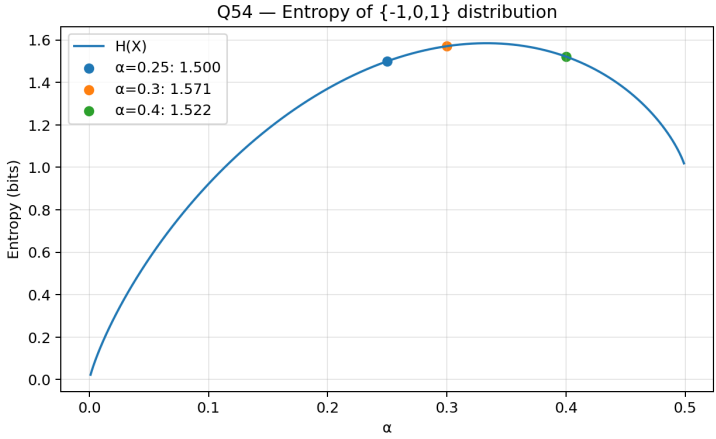


Fig. 1.2.31.1: Entropy $g\left(\alpha\right)$

1.2.32 Let $f\left(t\right)$ be periodic with fundamental period $T_0 > 0$, and $y\left(t\right) = f\left(\alpha t\right)$ with $\alpha > 1$. Their Fourier series are

$$f\left(t\right) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}\frac{2\pi}{T_0}kt} \tag{1.2.32.1}$$

$$y\left(t\right) = \sum_{k=-\infty}^{\infty} d_k e^{\mathrm{j}\frac{2\pi\alpha}{T_0}kt} \tag{1.2.32.2}$$

Which of the following is/are TRUE? (EC 2025)

a) $c_k = d_k$ for all k
b) $y(t)$ is periodic with fundamental period αT_0
c) $c_k = \frac{d_k}{\alpha}$ for all k
d) $y(t)$ is periodic with fundamental period $\frac{T_0}{\alpha}$

Solution: From (**??**),

$$y\left(t + \frac{T_0}{\alpha}\right) = f\left(\alpha t + T_0\right) \tag{1.2.32.3}$$

$$= y(t) \tag{1.2.32.4}$$

so the period is $\frac{T_0}{\alpha}$. From (**??**),

$$y(t) = \sum_k c_k e^{j\frac{2\pi k \alpha t}{T_0}} \tag{1.2.32.5}$$

$$\implies d_k = c_k \tag{1.2.32.6}$$

Hence (A) and (D) are true. The following code verifies the result in Fig. **??**.

codes/Fourier_Time_Scaling.py

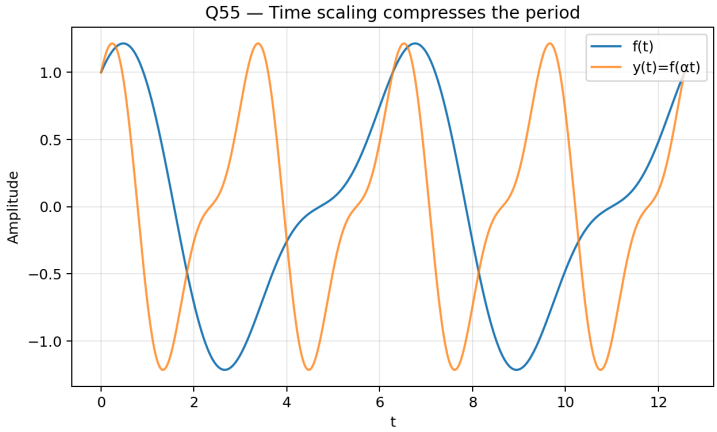


Fig. 1.2.32.1: $f(t)$ and $f(3t)$ have identical Fourier coefficients

1.2.33 Consider the system represented by the block diagram in Fig. **??**. Which of the signal flow graphs in Fig. **??** represent(s) this system? Choose the correct option(s). (EC 2025)

a) graph (A) b) graph (B) c) graph (C) d) graph (D)

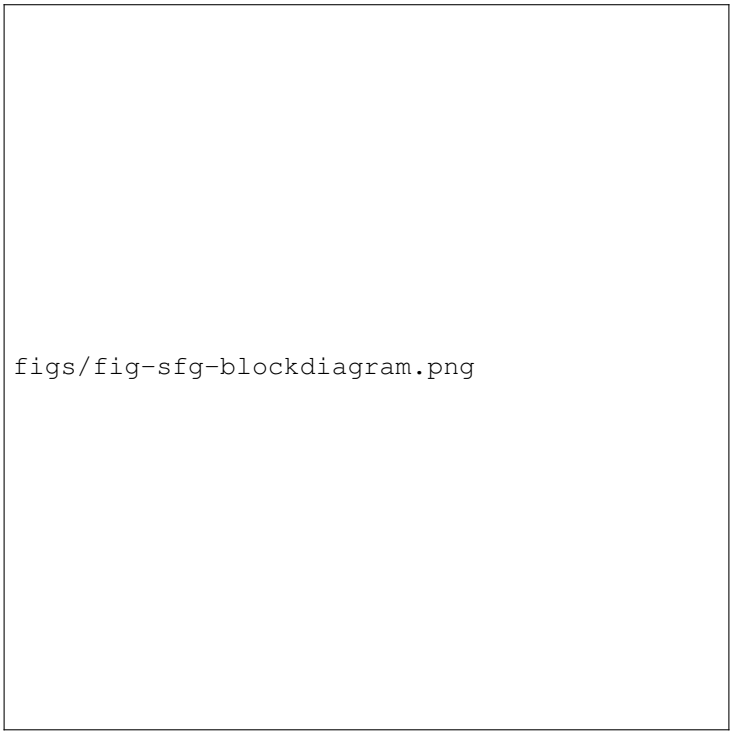


Fig. 1.2.33.1: Block diagram of the system

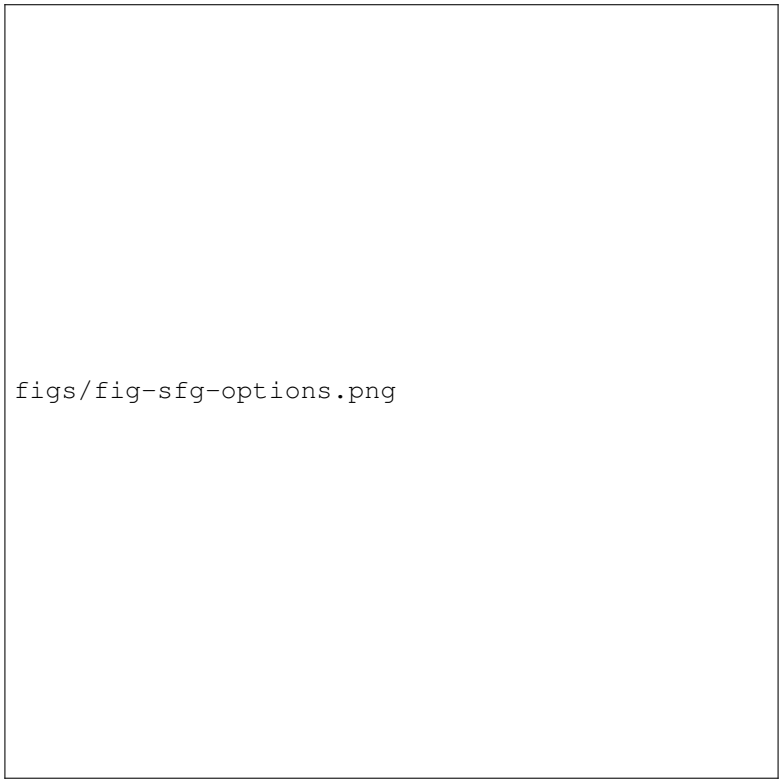


Fig. 1.2.33.2: Candidate signal flow graphs

Solution: From Fig. **??**,

$$e = R - G_1 e \tag{1.2.33.1}$$

$$\implies e = \frac{R}{1+G_1} \tag{1.2.33.2}$$

$$Y = (G_1 + G_2)\, e \tag{1.2.33.3}$$

$$\implies \frac{Y}{R} = \frac{G_1+G_2}{1+G_1} \tag{1.2.33.4}$$

From (**??**),

graph	loop gains	$\frac{Y}{R}$
(A)	$-G_1,\ -G_2$	$\frac{G_1+G_2}{1+G_1+G_2}$
(B)	$-G_1$	$\frac{G_1+G_2}{1+G_1}$
(C)	$+G_1,\ +G_2$	$\frac{G_1+G_2}{1-G_1-G_2}$
(D)	$-G_1,\ -G_2$	$\frac{G_1+G_2}{1+G_1+G_2}$

Only graph (B) represents the system. The following code verifies the solution in Fig. **??**.

codes/Mason_Signal_Flow_Graph.py

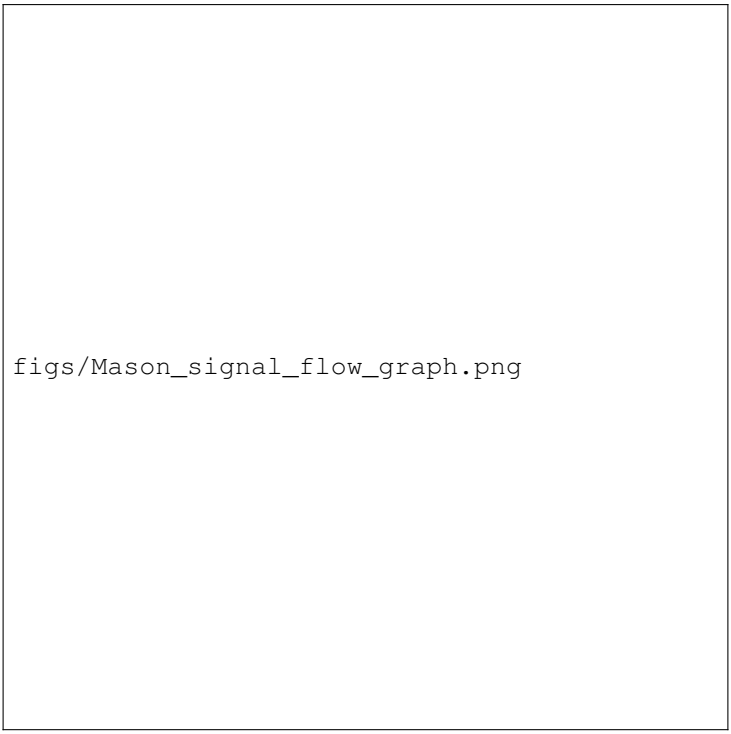


Fig. 1.2.33.3: Transfer functions of the four graphs

1.2.34 Two fair dice (faces $1, \ldots, 6$) are rolled. Let the random variable X be the sum of the outcomes. The expectation of X is _____ (rounded off to two decimal places). (EC 2025)

Solution: Let $X = X_1 + X_2$, where X_i is the outcome of die i. From (**??**),

$$E[X_i] = \frac{1+2+3+4+5+6}{6} \tag{1.2.34.1}$$

$$= 3.5 \tag{1.2.34.2}$$

$$E[X] = E[X_1] + E[X_2] \tag{1.2.34.3}$$

$$= 7.00 \tag{1.2.34.4}$$

The following code simulates the result in Fig. **??**.

codes/Two_Dice_PMF.py

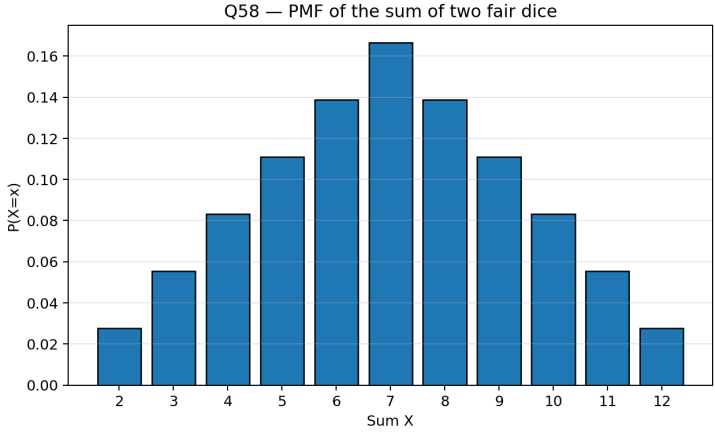


Fig. 1.2.34.1: PMF of the sum of two dice

1.2.35 Consider the vectors

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 3\sqrt{2} \end{pmatrix} \tag{1.2.35.1}$$

For a real scalar x, the value of $\min_x \|x\mathbf{a} - \mathbf{b}\|_2$ is _____ (rounded off to two decimal places), where $\|\mathbf{y}\|_2 = \sqrt{y_1^2 + y_2^2}$. (EC 2025)

Solution: From (**??**),

$$x^* = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} \tag{1.2.35.2}$$

$$= \frac{3\sqrt{2}}{2} \tag{1.2.35.3}$$

$$x^*\mathbf{a} - \mathbf{b} = \begin{pmatrix} \frac{3\sqrt{2}}{2} \\ -\frac{3\sqrt{2}}{2} \end{pmatrix} \tag{1.2.35.4}$$

$$\|x^*\mathbf{a} - \mathbf{b}\| = \sqrt{2 \cdot \frac{18}{4}} \tag{1.2.35.5}$$

$$= 3.00 \tag{1.2.35.6}$$

The following code plots the projection in Fig. **??**.

codes/Least_Squares.py

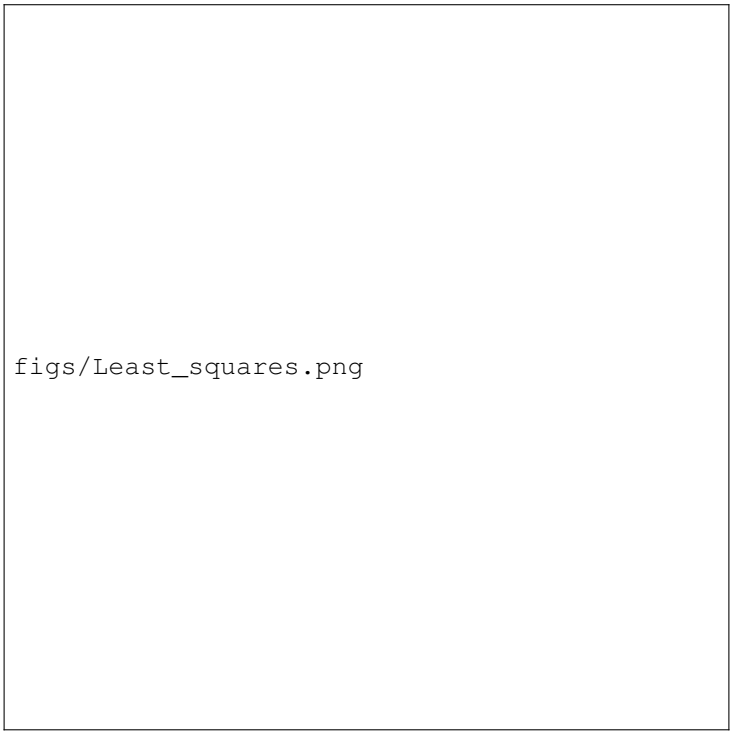


Fig. 1.2.35.1: Projection of $\mathbf{b}$ on the line through $\mathbf{a}$

1.2.36 X and Y are Bernoulli random variables on $\{0,1\}$ with joint pmf

$$\Pr(0,0)=0.06,\quad \Pr(0,1)=0.14,\quad \Pr(1,0)=0.24,\quad \Pr(1,1)=0.56 \tag{1.2.36.1}$$

The mutual information $I(X;Y)$ is _____ (rounded off to two decimal places). (EC 2025)

Solution: From (**??**),

$$\Pr(X=0)=0.06+0.14=0.20 \tag{1.2.36.2}$$
$$\Pr(X=1)=0.24+0.56=0.80 \tag{1.2.36.3}$$
$$\Pr(Y=0)=0.06+0.24=0.30 \tag{1.2.36.4}$$
$$\Pr(Y=1)=0.14+0.56=0.70 \tag{1.2.36.5}$$

Every joint probability equals the product of the marginals,

$$\Pr(X=0)\Pr(Y=0)=0.06 \tag{1.2.36.6}$$
$$\Pr(X=0)\Pr(Y=1)=0.14 \tag{1.2.36.7}$$
$$\Pr(X=1)\Pr(Y=0)=0.24 \tag{1.2.36.8}$$
$$\Pr(X=1)\Pr(Y=1)=0.56 \tag{1.2.36.9}$$

so from (**??**) and (**??**),

$$I\left(X;Y\right)=0.00 \tag{1.2.36.10}$$

The following code verifies the result in Fig. **??**.

codes/Mutual_Information_PMF.py

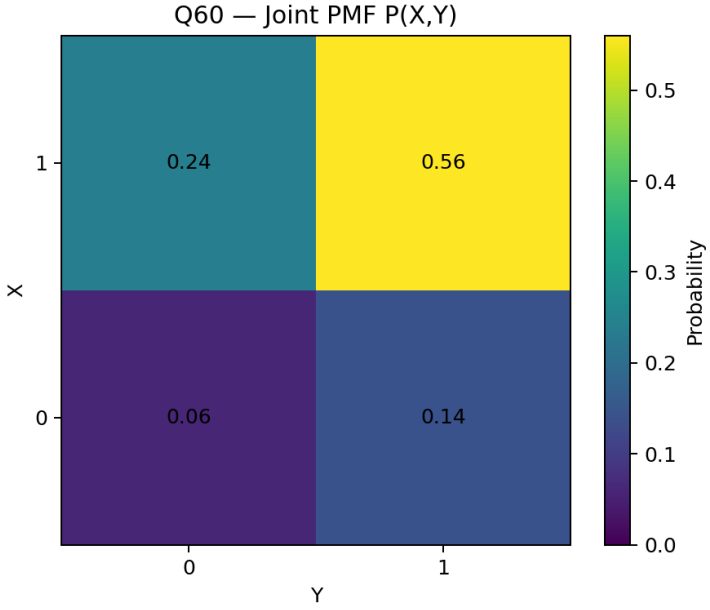


Fig. 1.2.36.1: Joint PMF against the product of the marginals

1.2.37 The diode in the circuit of Fig. **??** is ideal. The input voltage (in Volts) is $V_I = 10\sin\left(100\pi t\right)$, with t in seconds. The time duration (in ms, two decimal places) for which the diode is forward biased during one period of the input is _____. (EC 2025)

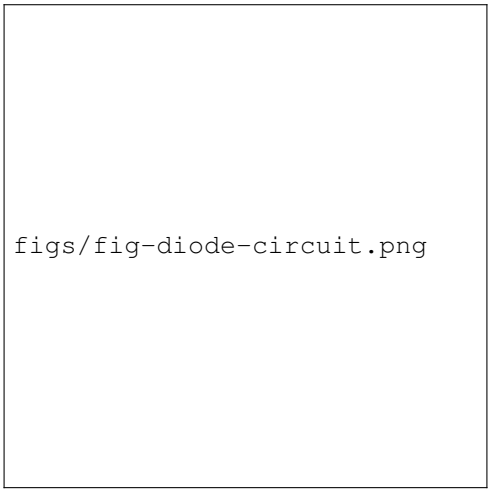


Fig. 1.2.37.1: Ideal diode circuit

Solution: From Fig. **??**, the cathode is at -5 V, so the diode conducts when $V_I > -5$ V. With $\theta = 100\pi t$,

$$10\sin\theta > -5 \tag{1.2.37.1}$$

$$\implies \sin\theta > -\frac{1}{2} \tag{1.2.37.2}$$

$$\Delta\theta = \frac{7\pi}{6} - \left(-\frac{\pi}{6}\right) \tag{1.2.37.3}$$

$$= \frac{4\pi}{3} \tag{1.2.37.4}$$

From (**??**),

$$\Delta t = \frac{\frac{4\pi}{3}}{100\pi} \tag{1.2.37.5}$$

$$= \frac{1}{75}\ \text{s} \tag{1.2.37.6}$$

$$= 13.33\ \text{ms} \tag{1.2.37.7}$$

1.2.38 In the circuit of Fig. **??**, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have set-up time 2 ns, hold time 0 ns and clock-to-Q delay 2 ns. The maximum clock frequency (in MHz, rounded off to the nearest integer) with no setup violation is _____. (EC 2025)

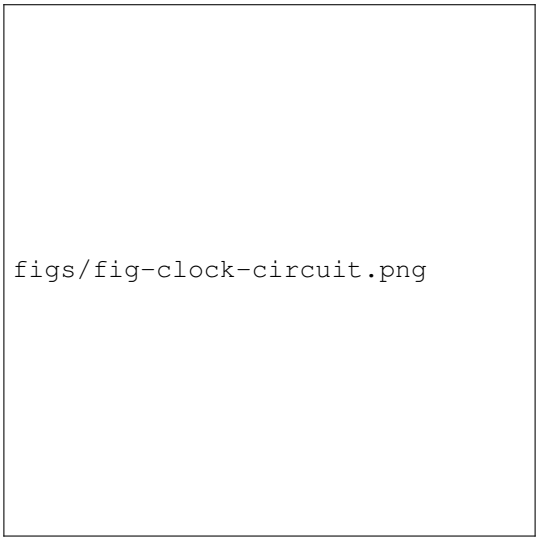


Fig. 1.2.38.1: Flip-flops with an AND gate in the data path

Solution: From Fig. **??**, the longest path is $FF_2 \rightarrow$ AND $\rightarrow FF_1$. From (**??**),

$$T_{\text{clk,min}} = t_{cq} + t_{AND} + t_{setup} \tag{1.2.38.1}$$
$$= 2 + 1 + 2 \tag{1.2.38.2}$$
$$= 5 \text{ ns} \tag{1.2.38.3}$$
$$f_{\max} = \frac{1}{5 \text{ ns}} \tag{1.2.38.4}$$
$$= 200 \text{ MHz} \tag{1.2.38.5}$$

The following code verifies the solution.

codes/Setup_Time_Verification.c

1.2.39 Two resistors are connected in a circuit loop of area 5 m^2, as in Fig. **??**. The loop lies in the x–y plane. A time-varying magnetic flux density $B(t) = 0.5t$ (in Tesla) is applied along the positive z-axis. The magnitude of the current I (in Amperes, two decimal places) in the loop is _____. (EC 2025)



Fig. 1.2.39.1: Circuit loop with two resistors

Solution: From Fig. **??**, $R_{\text{total}} = 4\ \Omega$. From (**??**),

$$|\mathcal{E}| = A\left|\frac{dB}{dt}\right| \tag{1.2.39.1}$$

$$= 5 \times 0.5 \tag{1.2.39.2}$$

$$= 2.5\ \text{V} \tag{1.2.39.3}$$

$$I = \frac{|\mathcal{E}|}{R_{\text{total}}} \tag{1.2.39.4}$$

$$= \frac{2.5}{4} \tag{1.2.39.5}$$

$$\approx 0.63\ \text{A} \tag{1.2.39.6}$$

The following code plots the flux, emf and current in Fig. **??**.

codes/Faraday_Loop_Current.py

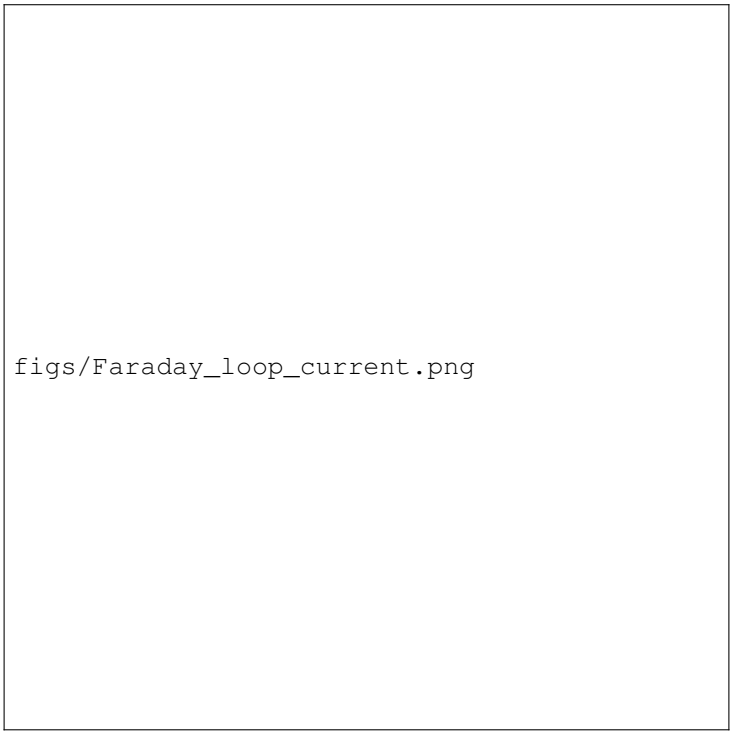


Fig. 1.2.39.2: Flux, emf and current in the loop